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This science textbook was created in compliance with the Ghana Standards Based Curriculum (NaCCA) and is intended to support efficient teaching and learning at the Basic 1 level.

This book has been thoughtfully crafted to improve young pupils' scientific knowledge, critical thinking, creativity, inquiry skills, and problem-solving ability. It encourages children to investigate their surroundings, ask questions, make discoveries, and cultivate positive attitudes toward science and scientific education.

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The approved teaching and learning criteria should be followed when using this textbook, which is intended solely for educational purposes. We hope that educators and students will effectively utilize this resource to enhance classroom instruction and cultivate valuable scientific experiences.

The information, exercises, and educational resources in this book have been thoughtfully created to be precise, age-appropriate, and pertinent to the requirements of the Basic 1 student.

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Foreword

Science helps us understand the world around us and enables us to make good decisions, solve problems, and use the resources in our environment responsibly. Introducing children to science at an early age therefore provides an important foundation for curiosity, creativity, critical thinking, and lifelong learning.

This Basic 1 Science (Agriculture) Textbook has been developed to support the teaching and learning of science in accordance with the Ghana Standards-Based Curriculum. It presents concepts in a simple, practical, and engaging manner, using examples and activities that relate closely to learners' everyday experiences.

The textbook encourages learners to observe, ask questions, explore, communicate ideas, and learn through practical activities. It connects science with the home, school, community, agriculture, health, food, and the natural environment, helping learners appreciate the importance of science in their daily lives.

It also supports the development of important competencies and values, including critical thinking and problem-solving, creativity and innovation, communication and collaboration, personal development, responsibility, respect, and care for the environment.

Teachers are encouraged to use the textbook creatively and adapt activities to suit the needs and circumstances of their learners. Parents and caregivers can further support learning by encouraging children to observe, ask questions, and discuss what they discover in their surroundings.

It is my hope that this textbook will inspire young learners to become curious, confident, responsible, and enthusiastic explorers of the world around them, while building a strong foundation for future learning in Science and Agriculture.

Introduction

Science helps us understand the world around us. Every day, we see, hear and experience many things that make us curious. We see plants growing, animals moving, the Sun shining, rain falling and many other interesting things happening around us.

Science helps us ask questions, observe things carefully and find answers. When we learn Science, we learn how things work and how we can use knowledge to solve problems in our homes, schools and communities.

In this book, you will explore different topics about living things, materials, the Earth, energy, the human body, technology and the environment. You will learn about the things around you and discover how Science is connected to your everyday life.

As a young scientist, you will learn to:

- observe things carefully;
- ask questions;
- investigate and discover new ideas;
- work with your classmates;
- carry out simple activities;
- explain what you have learned.

Science is not only about reading and remembering facts. It is also about exploring, thinking and discovering.

Always be curious. Ask questions. Observe carefully. Try new activities.

Remember:

Every great scientist begins by asking questions and wanting to learn more.
Welcome to the exciting world of science!

INTRODUCTION

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HOW TO USE THE BOOK

This Science book has been designed to help you learn through reading, observing, discussing and carrying out activities.

Each section contains simple explanations, examples and activities that will help you understand Science better.

As you use this book, look out for the following parts:

Let's Explore

This section encourages you to look closely at things around you. You may be asked to observe plants, animals, objects or events in your environment.

Example:

Look around your classroom. What things can you see?

Think and Talk

This section helps you share your ideas with your classmates. You will answer questions, discuss your thoughts and listen to the ideas of others. Remember that scientists learn by asking questions and sharing discoveries.

Activity

This section gives you opportunities to practise what you have learned. You may carry out simple investigations, make drawings, sort objects, observe changes or work with your classmates. Always follow instructions carefully and work safely.

Science Around Us

Science is connected to our daily lives. This section shows how Science helps us at home, school and in our communities.

You will discover how Science is used in farming, health, technology and caring for the environment.

Remember

This section highlights the important ideas you have learned. Read it carefully and use it to review your work.

Check Your Learning

At the end of each section, you will answer questions and complete activities to see how much you have learned. Do your best and ask your teacher for help when you need it.

SCIENCE SKILLS

Scientists use special skills to study and understand the world. In this book, you will practise these skills and become a better young scientist.

Observing

Observing means looking carefully at things around us and using our senses to gather information. We observe when we look at the colour, shape, size or features of an object.

For example: A scientist can observe a plant by looking at its leaves, stem and flowers.

Classifying

Classifying means grouping things according to their similarities and differences. Scientists classify things to make them easier to study.

For example, we can group things into:

- Living things
- Non-living things

Comparing

Comparing means looking at two or more things to find their similarities and differences.

For example: We can compare a bird and a fish. A bird can fly, while a fish swims.

Measuring

Measuring means finding the size, amount or quantity of something using suitable tools. We can measure Length, Height, Weight and Temperature. Scientists use measurements to make accurate observations.

Communicating

Communicating means sharing information, ideas and discoveries with others. Scientists communicate by speaking, writing, drawing and presenting their findings. You will practise communicating by discussing activities and sharing your observations with your classmates.

Predicting

Predicting means making a careful guess about what may happen. Scientists use what they already know to make predictions.

For example: A learner may predict that a plant without water will not grow well.

SCIENCE SKILLS

Investigating

Investigating means finding answers by asking questions, observing and carrying out activities. Scientists investigate to learn more about the world. You will carry out simple investigations throughout this book.

Creating

Creating means using ideas and materials to make something new. Scientists and inventors create tools and solutions to solve problems. You will use creativity during science activities and projects.

Remember:

Good scientists observe carefully, use evidence and make predictions based on what they know.

My Promise as a Young Scientist

I will:

- Ask questions.
- Observe carefully.
- Work safely.
- Respect living things and my environment.
- Share my ideas.
- Keep learning.

Science begins with curiosity. Let us explore and discover together!

STRAND 1: DIVERSITY OF MATTER

UNIT 1: LIVING AND NON-LIVING THINGS AROUND US

LESSON 1: THINGS AROUND US

Living Things and Non-Living Things



Everything around us is part of our environment. Our environment includes all the things we see around our homes, schools, communities and other places where we live. When we look carefully, we can find many different things such as plants, animals, water, soil, stones, houses, tables, books and vehicles.

The things around us can be grouped into two main groups: living things and non-living things.

Living things are things that are alive. They can grow, need food and water, breathe, move in their own way, and produce young ones. Plants and animals are examples of living things. A mango tree, a flower, a goat, a bird and a person are all living things.

Plants are living things because they grow and need water, air and sunlight to survive. Animals are also living things because they eat, breathe, grow and have

babies. Human beings are living things because we need food, water and air to stay alive.

Non-living things are things that are not alive. They do not grow, breathe, eat or produce young ones. Examples of non-living things include stones, chairs, books, pencils, tables, houses and cars.

Some non-living things are found naturally in our environment. These include rocks, water, soil and sand. Other non-living things are made by people to help us in our daily lives. These include school bags, clothes, houses and bicycles.

We can identify living and non-living things by observing their features. When we observe carefully, we can ask questions such as:

- Does it grow?
- Does it need food or water?
- Can it breathe?
- Does it produce young ones?

If the answer is yes, the thing may be living. If the answer is no, it is likely to be non-living.

Scientists also observe things around them carefully. Observation helps scientists learn about different things, compare them and group them according to their features.

Look Around!

Look around your classroom and your school environment.

- Can you find any living things?
- Can you find any non-living things?

Tell a friend about the things you see.

Discuss:

- What is the name of the thing?
- Is it living or non-living?
- Where did you find it?
- What helped you identify it?

Identifying Things in Our Environment

Our environment contains many different things. Some things are easy to see, while others may require careful observation. We can use our senses to identify things around us.

We use our eyes to see the colour, shape and size of things. We use our hands to feel whether something is rough, smooth, hard or soft. We use our ears to hear sounds, and we use our nose to identify some smells.

For example, when we see a tree, we can observe that it has leaves, branches and a stem. We know that the tree is a living thing because it grows and needs water and air.

When we see a chair, we can observe that it has legs and a seat. We know that the chair is a non-living thing because it does not grow or need food and water.

Different things can have different features. Some things are big, while others are small. Some are heavy, while others are light. Some are smooth, while others are rough. By observing these features, we can identify and group things correctly.

The place where we live affects the things we see around us. In a village, we may see farms, animals, trees and rivers. In a town or city, we may see buildings, roads, vehicles and shops. However, both places have living and non-living things.

Learning to identify things around us helps us understand and care for our environment.

Look Around!

Walk around your school with your teacher. Observe the things you see.

Find:

- Two plants
- Two animals
- Three objects made by people
- Two natural things

Discuss with your classmates:

- Which things are living?
- Which things are non-living?
- How did you know?

Activity: Let Us Observe

Go outside with your teacher and carefully observe your surroundings. Look for different living and non-living things.

Discuss:

- What things did you see?
- Which things are living?
- Which things are non-living?

- What features helped you identify them?
- Share your observations with your classmates.

After completing the activity, share your observations with your classmates. Explain why you grouped each thing as living or non-living.

Remember:

Observing carefully helps us discover more about the world around us.

Lesson 2: Plants and Animals Around Us

Living Things and Non-Living Things

Plants are living things that we see around us every day. They can be found in our homes, schools, farms, gardens, forests and communities. Plants are important because they provide us with food, medicine, materials for making things, and clean air for living.

Plants come in different shapes and sizes. Some plants are tall, while others are small. Some plants have large leaves, while others have small leaves. Some plants grow on land, while others grow in water.

In our locality, we can find many common plants. Some examples include maize, cassava, yam, plantain, mango trees, coconut trees, grass and flowers.

Farmers grow plants such as maize, cassava and yam because they provide food for people and animals. Trees such as mango and coconut are useful because they provide fruits, shade and materials that people use.

Flowers and other small plants make our surroundings beautiful. Grass grows in many places and provides food for some animals.

Plants have different parts that help them grow. The roots hold the plant firmly in the soil and take in water. The stem supports the plant and carries water to other parts. Leaves help the plant make food using sunlight. Flowers and fruits help some plants produce new plants.

We should take care of plants by watering them, protecting them from damage and planting more trees. Plants are important parts of our environment, and they help humans and animals survive.

Look Around!

Walk around your school or home with your teacher. Look carefully at the plants and animals around you.

Discuss:

- What plants did you see?
- What animals did you see?
- Where does it live?
- What does it eat?

Can you group the animals you found?

For example:

Animals with wings:

- Bird
- Butterfly

Animals with four legs:

- Goat
- Dog

Activity: Let Us Observe Plants and Animals

Go outside with your teacher and observe your surroundings. Find some plants and animals around you.

Discuss:

- Which plants did you see?
- Which animals did you see?
- Where did you find them?
- How are plants and animals important to us?

Share what you discovered with your classmates.

Remember:

Plants and animals are living things that make our environment beautiful and help us survive.



Lesson 3: What Living Things Need

Living things are all around us. Plants, animals and human beings are examples of living things. Although they are different from one another, all living things need certain things to survive, grow and stay healthy.

Living things need food, water, air and shelter. These things help them grow, get energy, protect themselves and continue living.

When we observe plants and animals around us, we can see that they depend on their surroundings for what they need. A plant in a garden needs water and sunlight to grow. A goat on a farm needs food, water and a safe place to rest. Human beings also need food, water, air and shelter to live.

Food

Food is one of the important things that living things need. Food gives living things energy to grow, move and carry out their daily activities. Human beings eat different kinds of food such as rice, maize, fruits, vegetables, fish and meat. These foods help our bodies grow strong and healthy.

Animals also need food to survive. Goats eat grass and leaves. Chickens eat grains and seeds. Some animals eat other animals for food.

Plants also need food, but they make their own food using sunlight, water and air. The food made by plants helps them grow and stay healthy.

Without food, living things become weak and may not grow properly.

Water

Water is another important need of living things. All plants and animals need water to survive. Human beings drink water to keep their bodies healthy. Animals also drink water to stay alive. Farmers provide water for animals on farms.

Plants need water to grow. Water helps plants carry nutrients from the soil to different parts of the plant. When plants do not get enough water, their leaves may dry up and the plant may die. We should use water carefully and avoid wasting it because living things depend on it.

Air

Air is found all around us. Living things need air to survive. Human beings and animals breathe in air through their bodies. Air helps them get the oxygen they need to stay alive.

Plants also need air to grow. They use air together with sunlight and water to make their own food. Although we cannot see air, we can feel it when the wind blows. Clean air is important for healthy living. We should help keep the air clean by avoiding activities that cause pollution.

Shelter

Shelter is a safe place where living things live and protect themselves. Human beings live in houses that protect them from rain, sun and cold weather. Animals also need places where they can rest and stay safe.

Examples of animal shelters include:

- Birds living in nests
- Dogs living in kennels
- Cows living in barns
- Fish living in water

Plants also need suitable places to grow. Some plants grow well in forests, gardens or farms where they can get enough sunlight, water and nutrients. Shelter helps living things stay protected from danger and harsh weather conditions.

Look Around!

Look around your home, school or community. Think about the living things you see. Choose one plant and one animal and answer these questions:

Plant:

- What is the name of the plant?
- Where does it grow?
- What does it need to grow?

Animal:

- What is the name of the animal?
- Where does it live?
- What does it need to survive?

Share your answers with your classmates.

Activity: Let Us Investigate

Work with your classmates and observe different living things around your school. Choose a plant and an animal.

Discuss:

- What does the plant need to grow?
- What does the animal need to survive?
- Where does each one get food?

- How are the plant and animal different?
- Where does each one get water?
- Where does each one live?

Discuss your answers with your classmates.

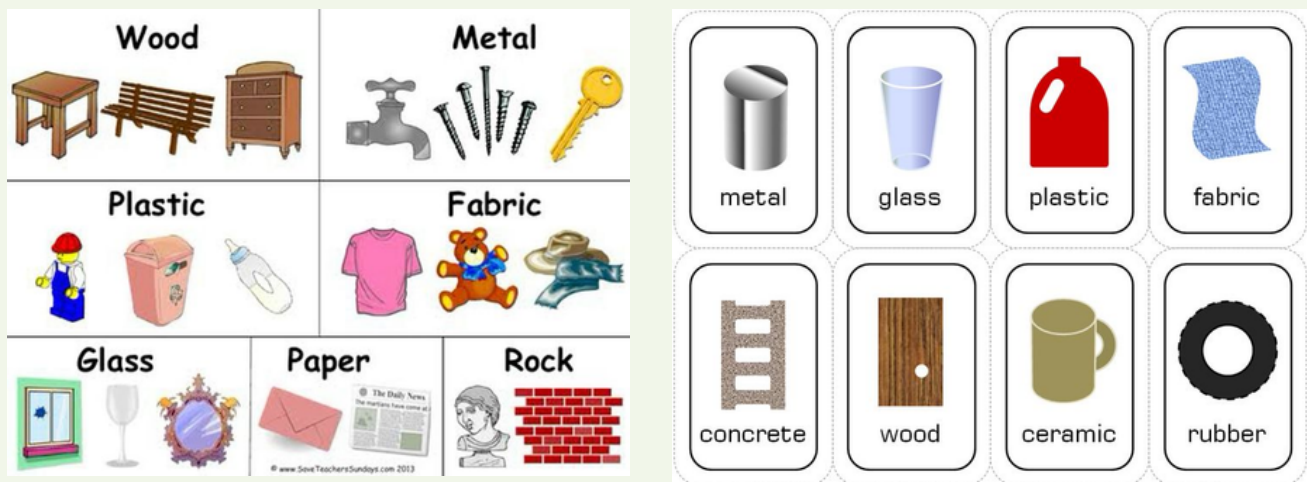
What did you discover about the needs of living things?

Remember!

All living things need food, water, air and shelter to survive. Taking care of plants, animals and our environment helps living things grow and remain healthy.

UNIT 2: MATERIALS AROUND US

LESSON 1: USEFUL MATERIALS IN OUR ENVIRONMENT



Everything around us is made from different materials. A material is what an object is made from. When we look around our homes, schools and communities, we can find many objects made from different materials.

For example, a chair may be made from wood or plastic. A spoon may be made from metal. A school uniform may be made from fabric, and a book may be made from paper.

Materials have different features that make them useful for different purposes. Some materials are strong, some are soft, some can bend, and some can last for a long time. People choose materials based on what they want to make.

Materials can come from nature or can be made by people. Wood comes from trees, metal comes from the Earth, and fabric can be made from materials such as cotton. Plastic is made by people and is used to make many things we use every day. Let us learn about some useful materials found in our environment.

Wood

Wood is a material that comes from trees. It is one of the most common materials used by people. Wood is strong and can be shaped into different objects.

Many things around us are made from wood. Examples include:

- Tables
- Chairs
- Doors
- Beds
- Pencils
- Wooden toys

Wood is useful because it is strong and easy to work with. People also use wood for building houses and making tools. We should take care of trees because trees provide wood and are important for our environment.

Metal

Metal is a strong material that comes from the Earth. It is used to make many objects because it is hard and can last for a long time.

Examples of things made from metal include:

- Cooking pots
- Spoons
- Nails
- Keys
- Bicycles
- Cars

Different types of metal are used for different purposes. Some metals are used because they are strong, while others are used because they do not easily break. We should handle metal objects carefully because some metal objects may have sharp edges or become hot.

Plastic

Plastic is a material made by people. It is commonly used because it is light, strong and can be made into many shapes. We can find plastic objects in our homes, schools and communities.

Examples include:

- Bottles
- Buckets
- Bags
- Rulers
- Toys
- Containers

Plastic is useful because it does not easily break and can hold different materials. However, plastic waste can harm the environment if it is not properly disposed of. We should keep our surroundings clean by reducing plastic waste and putting plastic items in the right places.

Fabric

Fabric is a soft material used for making clothes and other useful items. Fabric can be made from natural materials such as cotton or from materials made by people.

Examples of things made from fabric include:

- Clothes
- School uniforms
- Curtains
- Bags
- Blankets

Fabric comes in different colours, sizes and textures. Some fabrics are thick and strong, while others are thin and soft. Fabric helps to protect our bodies by making clothes that keep us comfortable.

Paper

Paper is a useful material that we use every day. It is usually made from trees.

Many things around us are made from paper, such as:

- Books
- Notebooks
- Newspapers
- Cards
- Paper bags

Paper helps us to write, draw and record information. Students use paper for learning activities in school. We should use paper carefully and avoid wasting it because trees are needed to produce paper.

Straw

Straw is a material that comes from plants. It is usually obtained from dry grasses or the stems of some crops.

People use straw to make different items such as:

- Baskets
- Mats
- Hats
- Roof coverings

Straw is useful because it is light and easy to shape. In some communities, people use straw materials to make traditional items.

We should appreciate materials from plants because they are useful resources from our environment.

Look Around!

Look around your classroom or home. Find objects made from different materials.

Can you find:

- Something made from wood?
- Something made from metal?
- Something made from plastic?
- Something made from fabric?
- Something made from paper?

Tell a friend:

- What is the object?
- What material is it made from?
- Why do you think that material was chosen?

Activity: Let Us Explore Materials

Go around your classroom with your teacher. Observe different objects and complete the table below:

Discuss:

- What object did you find?
- What material is it made from?
- Why do you think that material was chosen?
- Which materials did you find most often?
- Why do you think people use those materials?
- Which materials are hard, soft, strong, or flexible?
- Which materials can be reused or recycled?

Remember!

Materials are the things used to make objects around us.

Some useful materials include:

- Wood
- Metal
- Plastic
- Fabric
- Paper
- Straw

Different materials have different properties and uses. Some materials are strong, some are soft, some can bend, and some can break easily.

People choose materials based on what they need the object to do. For example, wood can be used to make tables and chairs because it is strong. Fabric is used to make clothes because it is soft and comfortable.

We should use materials wisely, reuse items when possible, and take care of our environment.

LESSON 2: WHERE MATERIALS COME FROM

Everything we use in our daily lives is made from different materials. These materials come from different sources around us. Some materials come from plants, some come from animals, and others come from the Earth.

People use these materials to make many useful things such as clothes, houses, tools, books and other objects we use every day. Learning where materials come from helps us understand how things are made and why we need to protect the resources in our environment.

Materials from Plants

Plants provide many useful materials. Trees, crops and other plants give us materials that people use to make different objects.

One important material that comes from plants is wood. Wood comes from trees and is used to make things such as tables, chairs, doors and pencils. Plants also provide materials such as cotton, which is used to make fabric for clothing. Cotton comes from cotton plants and is used to make shirts, dresses and other clothes.

Some plants provide materials used for making baskets, mats and other items. For example, straw from dry grasses and stems of plants can be used to make useful objects. Plants also give us materials used for making paper. Paper is made from trees and is used for books, notebooks, newspapers and other items.

Plants are very important because they provide materials and also help keep our environment healthy. We should protect plants by planting trees and avoiding unnecessary destruction of forests.

Examples of Materials from Plants

Plant Source	Material	Things Made From It
Tree	Wood	Tables, chairs, doors
Cotton Plant	Cotton Fabric	Clothes
Tree	Paper	Books, notebooks
Dry Grasses	Straw	Mats, baskets

Look Around!

Look around your home or school.
Can you find things made from plants?

Find objects such as:

- A wooden table
- A paper book
- A cotton cloth
- A straw basket

Tell a friend:

- What is the object?
- What material is it made from?
- Which plant did the material come from?

Materials from Animals

Animals also provide materials that people use to make useful things. Some materials come from the skin, hair or other parts of animals. One example is leather. Leather is made from animal skin and is used to make shoes, belts, bags and other items.

Animals such as sheep provide wool. Wool is a soft material that comes from the hair of sheep. It can be used to make warm clothes and blankets. Some animals also provide materials such as feathers. Feathers from birds can be used for decoration and making some useful items.

Animal materials have been used by people for many years. However, animals should always be treated carefully and protected.

Examples of Materials from Animals

Animal Source	Material	Things Made From It
Cow, goat or other animals	Leather	Shoes, bags, belts
Sheep	Wool	Clothes, blankets
Bird	Feathers	Decorations

Look Around!

Think about the things you use every day.

Can you find something that comes from an animal?

Discuss with your classmates:

- What is the object?
- What animal did the material come from?
- How is the object useful?

Materials from the Earth

The Earth provides many materials that people use. These materials are found in the ground, rocks, water and other parts of our environment. Some materials from the Earth include metals, stones, clay and sand.

Metals such as iron and aluminium come from the Earth. They are used to make objects such as cooking pots, tools, machines and vehicles.

Stones are found in many places and are used for building houses, roads and other structures.

Clay is a soft material found in the ground. People use clay to make pots, plates and other objects.

Sand is also found on the Earth. It is used for making glass and building materials. We must use materials from the Earth carefully because some resources can take a long time to replace.

Examples of Materials from the Earth

Earth Source	Material	Things Made From It
Ground	Metal	Tools, pots, vehicles
Rocks	Stone	Buildings, roads
Ground	Clay	Pots, plates
Beaches and rivers	Sand	Glass, buildings

Activity: Let Us Discover Where Materials Come From

Work with your classmates and observe different objects around your school.

Choose some objects and discuss:

- What is each object made from?
- Does the material come from a plant?
- Does it come from an animal?
- Does it come from the Earth?
- Where do people get this material from?
- Why do people use this material to make the object?
- What properties make the material useful?
- Can the material be reused or recycled?
- How can we use materials without harming the environment?

What did you discover about the materials we use every day?

Discuss your answers with your classmates.

- Which materials came from plants?
- Which materials came from animals?
- Which materials came from the Earth?

Remember!

Materials come from different sources.

Plants provide materials such as:

- Wood
- Paper
- Cotton
- Straw

Animals provide materials such as:

- Leather
- Wool
- Feathers

The Earth provides materials such as:

- Metals
- Stones
- Clay
- Sand

We should use materials wisely and protect the resources around us.

LESSON 3: STATES OF MATERIALS

Everything around us is made of materials. Materials can be found in different forms. Some materials are hard and keep their own shape, some can flow and take the shape of their containers, while others spread out and cannot be easily seen.

The different forms of materials are called states of materials. The three common states of materials are solids, liquids and gases. We can identify the state of a material by observing its shape, size and how it behaves.

Solids

A solid is a material that has a fixed shape and size. Solids do not flow from one place to another unless they are moved by someone or something.

Many things around us are solids. Examples include:

- Books
- Stones
- Tables
- Pencils
- Chairs
- Toys

A solid can be held with our hands because it keeps its own shape. For example, a book remains the same shape whether it is on a table, inside a bag or on the floor. Solids can have different features. Some solids are hard, while others are soft. Some are rough, while others are smooth.

For example:

- A stone is hard and rough.
- A pillow is soft.
- A glass cup is smooth.

We use solids to make many useful things around us. Wood is used to make furniture, metal is used to make tools, and plastic is used to make containers and toys.

Look Around!

Look around your classroom. Find five solid materials.

Tell a friend:

- What is the name of the object?
- What material is it made from?
- Is it hard or soft?
- Is it rough or smooth?

Liquids

A liquid is a material that can flow from one place to another. Unlike solids, liquids do not have a fixed shape. They take the shape of the container they are placed in.

Examples of liquids include:

- Water
- Milk
- Juice
- Cooking oil
- Rainwater

If we pour water into a cup, the water takes the shape of the cup. If we pour the same water into a bottle, it takes the shape of the bottle. Liquids can move and spread. This is why we must be careful when carrying liquids because they can spill.

Water is one of the most important liquids. Humans, animals and plants need water to survive. We use water for drinking, cooking, washing and growing plants. Some liquids are safe to use, while others can be harmful. We should never taste or touch unknown liquids.

Look Around!

Observe different liquids around you.

Can you find:

- Water?
- Juice?
- Cooking oil?

Discuss with your classmates:

- Where did you find the liquid?
- What container holds it?
- Does it take the shape of the container?

Gases

A gas is a material that spreads out and fills the space around it. Gases do not have a fixed shape or size.

Air is an example of a gas. Although we cannot see air, we know it is present because we can feel it when the wind blows.

Other examples of gases include:

- The gas used for cooking
- Air inside a balloon

Gases are all around us. Human beings and animals need oxygen in the air to breathe. Plants also need gases from the air to grow. A balloon becomes bigger when it is filled with air because the gas takes up space inside the balloon.

We should be careful when using gases because some gases can be dangerous if they are not handled properly.

Look Around!

Try this simple observation:

Take an empty balloon and blow air into it. What happens?

Discuss:

- Can you see the air?
- Does the balloon become bigger?
- What does this tell you about air?

Activity: Let Us Identify States of Materials

Work with your classmates and observe different materials around your school or home.

Look for examples of:

- Solids
- Liquids
- Gases

Discuss:

- Which materials are solids?
- Which materials are liquids?
- Which materials are gases?
- Where can you find them?

Remember!

Materials can exist in three common states:

Solids

- Have their own shape
- Can be held easily
- Examples: books, stones, chairs
- Do not flow like liquids
- Take up space and have a fixed size

Liquids

- Can flow
- Take the shape of their container

- Examples: water, milk, juice
- Do not have a fixed shape

Gases

- Spread out and fill spaces
- Cannot usually be seen
- Examples: air and cooking gas
- Do not have a fixed shape or size
- Can move freely and spread in all directions

Understanding the states of materials helps us learn more about the things around us. Materials can exist in different states, such as solids, liquids, and gases. Each state has its own properties and uses. By learning about these states, we can understand how materials change and how they are used in our daily lives.

LESSON 4: PROPERTIES OF MATERIALS

Materials around us have different features that help us identify and describe them. These features are called properties of materials. When we observe materials, we can look at how they feel, how they look, and how they behave when we touch or use them. Some materials are hard, while others are soft. Some materials are rough, while others are smooth. Some materials can bend or stretch, while others cannot.

Understanding the properties of materials helps us choose the right material for making different objects.

Hard and Soft Materials

Some materials are hard, while others are soft.

A hard material is a material that is strong and not easily pressed, broken or changed in shape. Hard materials are often used to make objects that need to last for a long time.

Examples of hard materials include:

- Stones
- Metals
- Wood
- Glass

A stone is hard because it does not easily change shape when we press it. Metal is also hard and is used to make tools, cooking pots and machines.

A soft material is a material that can be pressed, squeezed or changed more easily. Soft materials are usually comfortable to touch.

Examples of soft materials include:

- Cotton
- Sponge
- Cloth
- Foam

A pillow is soft because we can press it with our hands. A piece of cloth is also soft and can be used to make comfortable clothes.

Different materials are chosen for different purposes because of their hardness or softness. A chair may need hard wood or metal to support a person, while a pillow needs soft materials to make it comfortable.

Look Around!

Look around your classroom.

Find:

- One hard material
- One soft material

Touch each one carefully.

Tell a friend:

- What is the material?
- Is it hard or soft?
- What is it used for?

Rough and Smooth Materials

Materials can also be described by how their surfaces feel. The surface of an object is the part that we can touch. Some materials have rough surfaces. A rough material feels uneven or bumpy when we touch it.

Examples of rough materials include:

- Tree bark
- Sandpaper
- Stones
- Some types of wood

Some materials have smooth surfaces. A smooth material feels even and soft when we touch it.

Examples of smooth materials include:

- Glass
- Polished wood
- Plastic bottles
- Metal spoons

We use different materials because of how their surfaces feel. For example, a smooth spoon is comfortable to hold, while rough materials may be useful for objects that need more grip.

Look Around!

Find objects around you with different surfaces.

Can you find:

- Something rough?
- Something smooth?

Describe what you feel when you touch each object.

Discuss with your classmates:

- Which object feels rough?
- Which object feels smooth?
- How are they different?

Materials We Can Bend or Stretch

Some materials can change their shape when we push, pull or twist them. These materials can be bent or stretched. A material that can be bent changes shape when we apply force but may return to its original shape or keep the new shape.

Examples of materials that can bend include:

- Rubber
- Thin plastic
- Wire
- Some types of wood

A material that can stretch becomes longer when we pull it.

Examples of materials that can stretch include:

- Rubber bands
- Elastic materials in clothes
- Balloons

Some materials cannot bend or stretch easily. For example, a stone and a glass cup can break if we try to bend them.

The ability of a material to bend or stretch makes it useful for different purposes. Rubber bands are useful because they can stretch and return to their original shape. Elastic materials are used in clothing because they can expand and fit different body sizes.

Look Around!

Find materials around you that can bend or stretch.

Try to observe:

- A rubber band
- A piece of cloth
- A plastic ruler

Discuss:

- Which material can bend?
- Which material can stretch?
- Which material cannot change shape easily?

Activity: Let Us Test Materials

Work with your classmates and observe different materials in your classroom. Choose different objects and examine their properties. Discuss what each material looks and feels like, and explain why it is suitable for making that object.

Discuss:

- Which materials were hard?
- Which materials were soft?
- Which materials are rough?
- Which materials are smooth?
- Which materials could bend or stretch?
- Which materials could absorb water or keep water out?
- Which materials were heavy or light?
- Why are different materials useful for different purposes?

Share your findings with your classmates.

Remember!

Materials have different properties.

Hard materials:

- Do not easily change shape
- Are usually strong and difficult to break
- Examples: stones, metals, wood

Soft materials:

- Can be pressed or squeezed
- Are easy to change or cut
- Examples: sponge, cotton, cloth

Rough materials:

- Have uneven surfaces
- Feel bumpy when touched
- Examples: stones, tree bark

Smooth materials:

- Have even surfaces
- Feel flat and slippery when touched
- Examples: glass, plastic

Materials that can bend or stretch:

- Can change shape when force is applied
- Can return to their original shape after bending or stretching
- Examples: rubber bands and elastic materials

Knowing the properties of materials helps us understand why different materials are used to make different things. It also helps us choose the right materials for different purposes and use them safely and wisely.

STRAND 2: PATTERNS IN NATURE

UNIT 3: EARTH SCIENCE

LESSON 1: WEATHER AROUND US



Weather describes the condition of the air and sky around us at a particular time. When we look outside, we may notice that the day can be bright and hot, cloudy, rainy or windy. These changes in the weather affect the activities we do every day.

Weather helps us decide what clothes to wear, what activities we can do and how we take care of our surroundings. Farmers also observe the weather because it helps them know when to plant crops and take care of their farms.

We can observe the weather by looking at the sky, feeling the temperature, noticing rainfall and watching how the wind moves things around us. There are different types of weather that we experience in our environment. Some common types are sunny weather, rainy weather and windy weather.

Sunny Weather

Sunny weather happens when the Sun shines brightly in the sky. During sunny weather, the sky is usually clear, and there may be little or no clouds. On a sunny day, we may feel heat from the Sun. People often wear light clothes and use hats or umbrellas to protect themselves from too much sunlight.

The Sun is important because it gives us light and heat. Plants need sunlight to grow, and people use sunlight for activities such as drying clothes and drying some foods.

However, staying under the Sun for too long can make us feel uncomfortable. We should protect ourselves by staying in shaded areas, wearing protective clothing and drinking enough water.

Examples of activities we can do during sunny weather include:

- Playing outside
- Drying clothes
- Drying crops
- Working in the garden

Ask yourself:

- Is the Sun shining?
- Is the sky bright or cloudy?
- Can you feel the heat from the Sun?

Tell your classmates what the weather looks like today.

Rainy Weather

Rainy weather happens when water falls from clouds in the sky as rain. Rain is an important part of nature because plants, animals and humans need water to survive.

During rainy weather, the sky may become dark with clouds. We may see raindrops falling, hear the sound of rain and notice water collecting on the ground.

Rain helps plants grow because it provides water for crops, trees and flowers. Farmers depend on rainfall to grow many crops. People need to prepare for rainy weather. We can use umbrellas and raincoats to keep ourselves dry. We should also avoid walking through dirty water because it may contain germs that can make us sick.

Examples of activities affected by rainy weather include:

- Farming
- Outdoor games
- Travelling
- Drying clothes

Look Around!

Observe what happens when it rains.

Discuss with your classmates:

- What changes do you notice in the sky?
- What happens to the ground?
- How do people protect themselves from rain?

Windy Weather

Windy weather happens when air moves from one place to another. We cannot see air, but we can feel it when the wind blows. During windy weather, we may see leaves moving, trees shaking, clothes drying faster and objects such as flags waving.

Wind can be gentle or strong. A gentle wind can make us feel comfortable, while a strong wind can sometimes damage houses, trees or other objects.

Wind is useful in many ways. It helps to move clouds, dry clothes and can be used to produce energy.

Examples of things we may notice during windy weather include:

- Moving leaves
- Flying kites
- Waving flags
- Moving clouds

Look Around!

Go outside with your teacher on a windy day. Observe your surroundings.

Can you see:

- Leaves moving?
- Trees shaking?
- Flags waving?

Discuss:

- How do you know that the wind is blowing?
- What things around you are affected by the wind?

Activity: Let Us Observe the Weather

Work with your classmates to observe the weather each day for one week. Look carefully at the sky and your surroundings.

Discuss:

- Is the weather sunny, rainy, cloudy or windy?
- What changes do you notice?
- Which type of weather did you observe most often?
- What activities can you do during sunny weather?
- Why is rain important?
- How can you tell when the wind is blowing?

Share your observations with the class.

Remember!

Weather tells us what the air and sky are like at a particular time. The three common types of weather are:

Sunny weather

- The Sun shines brightly.
- The day is usually warm.

Rainy weather

- Water falls from clouds.
- Plants and animals get water.

Windy weather

- Air moves around us.
- We can see its effects on objects.

Observing the weather helps us understand our environment and make good decisions every day.

LESSON 2: OBSERVING WEATHER CHANGES



Weather changes from time to time. A day that begins with bright sunshine may later become cloudy or rainy. By observing the weather regularly, we can understand the changes that happen in our environment.

People observe weather changes for different reasons. Farmers observe the weather to know when to plant crops and care for their farms. Pilots, drivers and travellers also pay attention to weather conditions to plan their activities safely.

We can observe weather by looking at the sky, feeling the air around us and noticing changes in our surroundings. We can ask questions such as:

- Is the sky sunny or cloudy?
- Is it raining?
- Is the wind blowing?
- Is the day hot or cool?

Recording what we observe helps us remember and compare changes in the weather.

Keeping Weather Records

A weather record is a way of writing down information about the weather we observe each day. Weather records help us know how the weather changes over time. We can keep records for a few days, weeks or even months. To keep a weather record, we observe the weather and write down what we see.

For example, we can record whether the day is:

- Sunny
- Rainy
- Windy
- Cloudy

We can also record other observations such as:

- How hot or cool the day feels
- How much rain falls
- What happens to plants and animals during different weather conditions

A simple weather chart can help us keep our records.

Example:

Day	Weather Condition	Observation
Monday	Sunny	The sky was bright and clear.
Tuesday	Rainy	Water fell from the clouds.
Wednesday	Windy	Leaves moved on the trees.

By keeping weather records, we learn to notice patterns in the weather. For example, we may discover that some days are usually rainy while others are usually sunny.

Look Around!

Observe the weather outside your classroom today.

Talk with your classmates:

- What does the sky look like?
- Is the Sun shining?
- Is there rain?
- Can you feel the wind?

Record your observation in a weather chart.

Weather Instruments

Scientists use special tools called weather instruments to observe and measure weather conditions. Weather instruments help us collect accurate information about the weather.

Some common weather instruments include:

Rain Gauge

A rain gauge is an instrument used to measure the amount of rainfall. It collects rainwater and helps us know how much rain has fallen. Farmers and weather scientists use rain gauges to study rainfall.

Thermometer

A thermometer is an instrument used to measure temperature. It helps us know how hot or cold the air is.

For example:

- A high temperature means the weather is hot.
- A low temperature means the weather is cooler.

Wind Vane

A wind vane is an instrument used to show the direction from which the wind is blowing. It helps people know whether the wind is coming from the north, south, east or west.

Anemometer

An anemometer is an instrument used to measure the speed of the wind. It helps scientists know whether the wind is moving slowly or quickly.

Weather Symbols

People also use symbols to record weather conditions.

Examples:

 Sunny weather

 Rainy weather

 Cloudy weather

 Windy weather

Weather symbols make it easier for children and adults to understand weather information quickly.

Look Around!

Look at the weather instruments shown by your teacher.

Discuss:

- What is the name of the instrument?
- What does it measure?
- How does it help us understand weather?

Activity: Let Us Keep a Weather Record

Work with your classmates to observe the weather for one week. Talk about the weather you observed.

After one week, discuss your findings

- Which weather condition happened most often?
- Which day had the most rain?
- Did the weather change during the week?
- What signs helped you identify the weather?

Remember!

We can observe weather changes by carefully watching our environment. Keeping weather records helps us:

- Understand weather patterns

- Compare different weather conditions
- Plan our activities

Weather instruments help us measure weather conditions.

Examples:

- Rain gauge → measures rainfall
- Thermometer → measures temperature
- Wind vane → shows wind direction
- Anemometer → measures wind speed

Observing and recording weather helps us learn more about the world around us.

UNIT 4: LIFE CYCLES OF ORGANISMS

LESSON 1: GROWING AND CHANGING



Living things do not stay the same throughout their lives. They grow, change and develop as time passes. Plants, animals and human beings all go through stages of growth.

Growth means an increase in size and development of living things. As living things grow, they may change in appearance, shape and abilities.

A small plant does not look the same as a fully grown plant. A baby animal does not look or behave exactly like an adult animal. These changes are part of the life cycle of living things. By observing plants and animals around us, we can learn how they grow and change.

Growth in Plants

Plants are living things that grow and change. Most plants begin their life from a seed.

A seed contains a small baby plant that can grow when it gets the right conditions. When a seed is planted in suitable soil and receives water, air and sunlight, it begins to grow.

The first stage of plant growth is called germination. During germination, a small root begins to grow from the seed. A young shoot then grows upward and develops into a small plant.

As the plant continues to grow, it develops more roots, stems and leaves. The plant becomes bigger and stronger.

A simple plant growth process is:

Seed → Young plant → Mature plant → Flower or fruit

Different plants grow in different ways. Some plants grow into large trees, while others remain small. Some plants produce flowers and fruits, while others produce seeds in different ways.

Examples of plants that grow around us include:

- Maize
- Beans
- Mango trees
- Flowers
- Grass

Plants need certain things to grow well. These include:

- Water
- Air
- Sunlight
- Nutrients from the soil

Without these things, plants may become weak or stop growing. We can help plants grow by watering them, protecting them from damage and keeping their surroundings clean.

Look Around!

Observe plants around your home or school.

Choose one plant and discuss:

- What does the plant look like?
- Is it small or big?
- Does it have leaves?
- Does it have flowers or fruits?

Compare a young plant with a bigger plant. What differences can you see?

Growth in Animals

Animals are also living things that grow and change. Animals begin life as young ones and gradually develop into adults.

Different animals grow in different ways. Some animals look similar to their parents as they grow. For example, a baby goat looks like a smaller version of an adult goat. As it grows, its body becomes bigger and stronger.

Some animals change greatly as they grow. For example, a butterfly begins life as an egg, changes into a caterpillar, then becomes a butterfly. Animals go through different stages of growth. These stages are called the life cycle.

Examples:

Chicken:

Egg → Chick → Adult chicken

Goat:

Baby goat → Young goat → Adult goat

Butterfly:

Egg → Caterpillar → Butterfly

Animals need food, water, air and shelter to grow properly. Young animals need care and protection until they become adults. People help animals grow by providing them with proper food, clean water and safe places to live.

Look Around!

Observe animals around your community.

Think about:

- Have you seen a baby animal?
- How is it different from an adult animal?
- What does the animal need to grow?

Share your answers with your classmates.

Activity: Let Us Observe Growth

Work with your classmates to discuss how living things grow and change. Look at the examples in this lesson.

Discuss:

- How does a plant begin its life?
- How does a chicken change as it grows?
- How does a butterfly change as it grows?
- How does a frog change as it grows?
- How is a young animal different from an adult animal?
- Why do living things need care while growing?

Share your ideas with the class.

Remember!

All living things grow and change.

Plants:

- Begin as seeds
- Grow into bigger plants
- Produce flowers, fruits or seeds

Animals:

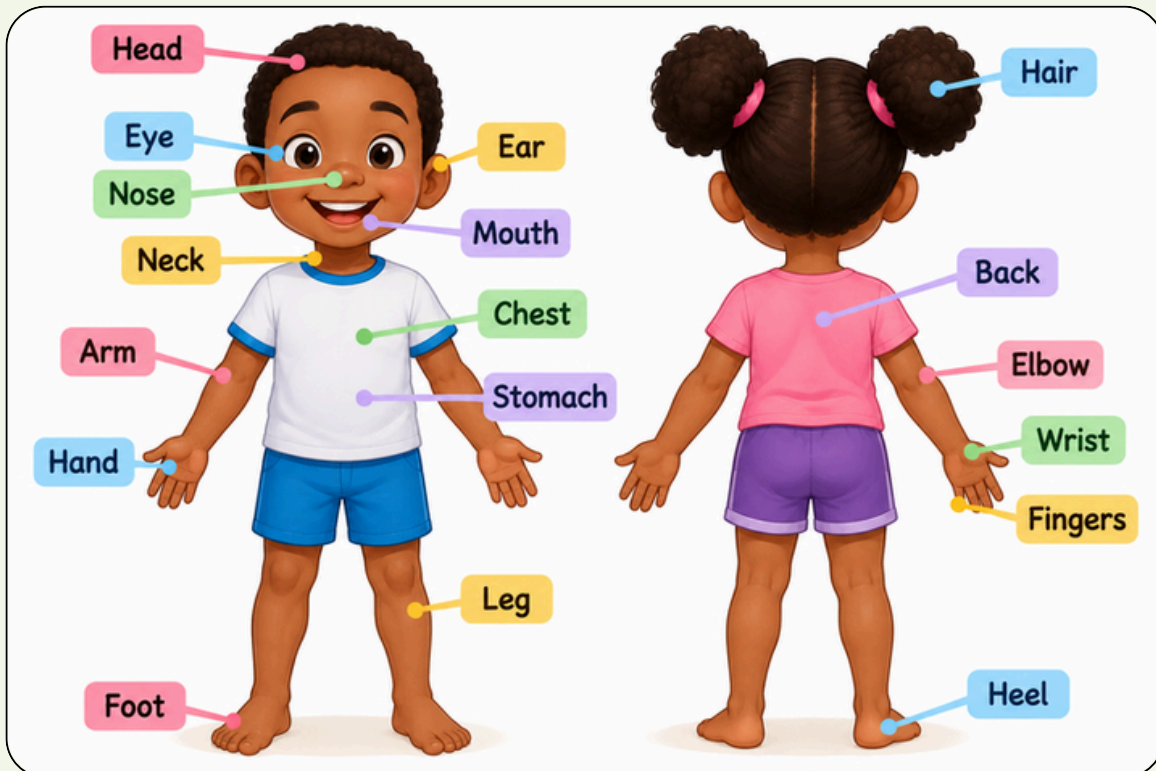
- Begin as young ones
- Grow into adults
- Change in size and appearance

Plants and animals need proper care, food, water and a suitable environment to grow healthy.

STRAND 3: DIVERSITY OF MATTER

UNIT 5: THE HUMAN BODY SYSTEMS

LESSON 1: PARTS OF MY BODY



Our body is a wonderful system made up of many different parts. Each part of our body has a special job that helps us live, move, learn and interact with the world around us.

We use our body every day when we walk, run, play, eat, talk, see and touch things. By learning about our body parts, we understand how to take good care of ourselves and stay healthy.

The human body has many parts. Some parts can be seen from outside our body. These are called external body parts.

External Body Parts

External body parts are the parts of our body that we can see and touch. These parts help us perform different activities every day.

Some common external body parts include:

- Head
- Eyes
- Ears
- Nose
- Mouth
- Neck
- Arms
- Hands
- Fingers
- Legs
- Feet
- Skin

Each body part has a special function that helps our body work properly.

The Head

The head is the upper part of our body. It contains important organs such as the brain, eyes, ears, nose and mouth. The brain helps us think, learn and control our body movements.

The Eyes

Our eyes help us to see things around us. We use our eyes to see colours, shapes, sizes and objects.

For example, we use our eyes to read books, watch people and find our way around places. We should protect our eyes by avoiding looking directly at very bright lights and keeping them clean.

The Ears

Our ears help us to hear sounds. We use our ears to listen to people speaking, hear music and notice sounds around us. We should take care of our ears by keeping them clean and avoiding putting sharp objects inside them.

The Nose

Our nose helps us to smell different things. It also helps us breathe in air. We use our nose to identify smells such as the smell of flowers, food or smoke.

The Mouth

The mouth helps us to eat food, drink water and speak. Inside the mouth are teeth that help us chew food. The tongue helps us taste food and move it around our mouth.

The Hands and Fingers

Our hands help us hold, carry and move objects.

Our fingers help us do many activities such as writing, drawing, picking things up and playing games.

The Arms

Our arms help us lift, carry and move things. We use our arms when we wave, throw objects or hug someone.

The Legs and Feet

Our legs help us stand, walk, run and jump. Our feet help us balance and move from one place to another.

The Skin

The skin covers our whole body. It protects the inside parts of our body. Our skin also helps us feel things such as heat, cold, rough surfaces and smooth surfaces.

Functions of Body Parts

Every body part has a special function. When all our body parts work together, we can perform different activities.

Examples:

Body Parts	Function
Eyes	Help us see
Ears	Help us hear
Nose	Helps us smell and breathe
Mouth	Helps us eat, drink and speak
Hand	Help us hold and touch thing
Legs	Help us walk and run
Feet	Help us balance and move
Skin	Protects the body and helps us feel

Our body parts work together to help us live and enjoy our daily activities.

For example, when we eat food:

- Our eyes help us see the food.
- Our hands help us hold the food.
- Our mouth helps us chew and swallow the food.

When we play a game:

- Our eyes help us see.
- Our legs help us run.

Our hands help us hold or throw objects.

Look Around!

Look at yourself and your classmates.
Identify some external body parts.

Discuss:

- What is the name of the body part?
- Where is it located?
- What does it help us do?

Activity: Let Us Identify Body Parts

Look at the pictures of the body parts. With your teacher, identify:

- Eyes
- Hands
- Legs
- Ears
- Nose

Discuss:

- What does each body part help us do?
- Where is each body part found?

Activity: Exploring Our Body

Look at the picture of the human body. With your teacher, identify:

- Head
- Eyes
- Nose
- Mouth
- Hands
- Legs
- Feet

Discuss:

- Which body parts help us see?
- Which body parts help us walk?
- Which body parts help us smell?
- Which body parts help us hold things?

Remember!

Our body is made up of many parts that work together. External body parts are the parts we can see and touch.

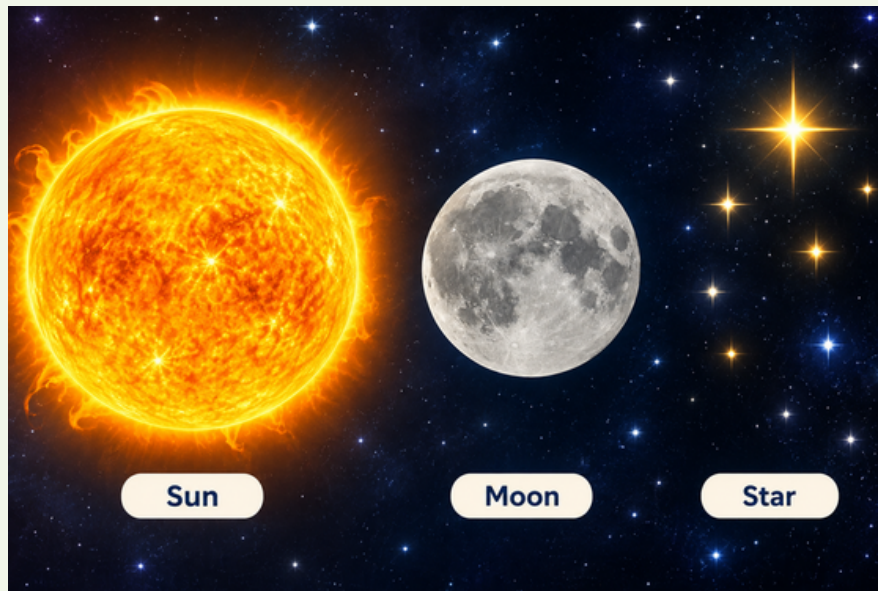
Examples include:

- Eyes
- Ears
- Nose
- Hands
- Legs
- Feet

Each body part has an important function. Taking care of our bodies helps us stay healthy and active.

UNIT 6: THE SOLAR SYSTEM

LESSON 1: OBJECTS WE SEE IN THE SKY



When we look up at the sky during the day or night, we can see many interesting objects. Some objects in the sky give us light, while others reflect light from the Sun.

The sky changes at different times of the day. During the day, we usually see the bright Sun shining above us. At night, we may see the Moon and many stars.

The objects we see in the sky are part of the natural world around us. They help us learn more about our planet and the space beyond it.

Some common objects we see in the sky are:

- The Sun
- The Moon
- Stars

The Sun

The Sun is a very large star that gives us light and heat. It is the brightest object we see in the sky during the day.

The Sun is important to all living things. Plants need sunlight to grow and make their own food. Human beings and animals also depend on the Sun because it provides warmth and helps us see during the day.

We use sunlight for many activities, such as:

- Drying clothes
- Drying food crops
- Providing light during the day

The Sun appears to move across the sky during the day. In the morning, we see it rising in the east, and in the evening, we see it setting in the west.

Although the Sun looks small from where we are, it is actually very large. It is far away from the Earth, which is why it appears smaller in the sky.

We should protect our eyes and avoid looking directly at the Sun because its bright light can harm our eyes.

Look Around!

Observe the sky during the day.

Discuss with your classmates:

- Where can you see the Sun?
- What does the Sun provide for us?
- How does sunlight help plants grow?

The Moon

The Moon is an object we often see in the sky at night. Unlike the Sun, the Moon does not produce its own light. It reflects light from the Sun, which makes it appear bright.

The Moon changes its appearance at different times. Sometimes we see a full round Moon, while at other times we see only part of it.

The Moon is important because it helps us understand changes that happen in the sky. People have observed the Moon for many years to learn about time and seasons.

The Moon does not have its own air, water or living things like the Earth. It is a natural object that moves around the Earth.

Look Around!

Look at the sky at night with an adult. Observe the Moon.

Discuss:

- What shape does the Moon look like?
- Is it bright or dark?
- Where can you see it in the sky?

Stars

Stars are bright objects that we see in the sky, especially at night. Like the Sun, stars produce their own light. There are many stars in the sky, but they look small because they are very far away from Earth.

Some stars appear brighter than others. On a clear night, we can see many stars shining in the sky. People have used stars for many years to help them find directions when travelling. Scientists also study stars to learn more about space.

The Sun is also a star, but it appears larger and brighter because it is much closer to Earth than other stars.

Look Around!

On a clear night, observe the sky.

Discuss with your family or classmates:

- How many stars can you see?
- Are all stars the same brightness?
- How do stars make the night sky look?

Activity: Let Us Observe the Sky

Work with your classmates and observe the sky at different times.

Discuss:

- What can you see in the sky during the morning?
- What can you see during the afternoon?
- What can you see at night?
- Which objects can be seen during the day?
- Which objects can be seen at night?
- Why is the Sun important to living things?

Activity: Exploring Objects in the Sky

Look carefully at the pictures of the Sun, Moon and stars. With your teacher, identify each object.

Discuss:

- Which object gives us light and heat?
- Which object reflects light from the Sun?
- Which objects produce their own light?
- Which objects can we usually see at night?

Remember!

The sky contains many interesting objects.

The Sun

- Gives us light and heat.
- Helps plants grow.
- Gives us light during the day.

The Moon

- Is seen mostly at night.
- Reflects light from the Sun.
- Changes its shape as we see it from Earth.

Stars

- Produce their own light.
- Can be seen clearly at night.
- Look small because they are very far away.

Observing objects in the sky helps us learn more about our world and the space around us.

UNIT 7: ECOSYSTEM

LESSON 1: LIVING THINGS AND THEIR ENVIRONMENT

Living things are found everywhere around us. Plants, animals and human beings live in places where they can get the things they need to survive, such as food, water, air and shelter.

The place where a living thing lives is called its environment. An environment provides the conditions that help living things grow, survive and reproduce.

Different plants and animals live in different environments. Some live on land, some live in water, and some live in places with many trees and plants. The environment where a living thing lives affects how it grows and survives.

Living things depend on their environment, and the environment also depends on living things. Plants provide food and shelter for some animals, while animals help plants in different ways.

Places Where Plants and Animals Live

Plants and animals live in different places called habitats. A habitat is a place where a living thing finds the food, water, air and shelter it needs to survive.

Different habitats have different conditions. Some places are wet, some are dry, some have many trees, and some have large amounts of water.

Forest

A forest is a place with many trees, plants and other living things. Many animals live in forests because they can find food and shelter there.

Examples of animals found in forests include:

- Birds
- Monkeys
- Insects
- Snakes

Plants found in forests include:

- Large trees
- Flowers
- Bushes

Trees provide homes for animals and help keep the environment healthy.

Farm

A farm is a place where people grow plants and keep animals.

Farm plants include:

- Maize
- Cassava
- Beans
- Vegetables

Farm animals include:

- Goats
- Chickens
- Sheep
- Cows

Farms provide food for people and animals.

Water Habitats

Some living things live in water. Rivers, ponds, lakes and oceans are examples of water habitats.

Animals that live in water include:

- Fish
- Frogs
- Other small water animals

Some plants also grow in water. Water provides these living things with the conditions they need to survive.

Grasslands

Grasslands are places where grasses grow in large amounts. Many animals live in grasslands because they can find food and space to move.

Examples of animals found in grasslands include:

Cattle

- Goats
- Birds
- Insects

Homes and Communities

Some plants and animals live close to human beings.

Examples include:

- Flowers growing around houses
- Trees planted in school compounds
- Dogs and cats living with families

People also depend on these living things for different purposes.

Look Around!

Look around your school or community. Can you find places where plants and animals live?

Discuss:

- What plants can you see?
- What animals can you see?
- Where do they live?

Tell your classmates about one habitat you observed.

Relationships Between Living Things and Their Surroundings

Living things depend on their surroundings to survive. Their surroundings provide the things they need, such as food, water, air and shelter. Plants and animals are connected to each other and to their environment.

Plants and Their Environment

Plants need sunlight, water, air and nutrients from the soil to grow.

For example:

- A tree uses sunlight to make food.
- Roots take in water from the soil.
- Leaves use air to help the plant grow.

Plants also provide benefits to other living things.

Plants provide:

- Food for humans and animals
- Shelter for some animals
- Oxygen that living things need to breathe

Animals and Their Environment

Animals depend on their environment for survival.

Animals need:

- Food
- Water
- Air
- Shelter

For example:

- Birds build nests in trees.
- Fish live in water.
- Goats live on farms and eat plants.

Animals also affect their environment. Some animals help plants by spreading seeds or helping flowers produce new plants.

Human Beings and the Environment

Human beings depend on the environment for many things.

We get:

- Food from plants and animals
- Water from rivers and other sources
- Materials for making houses and tools

People also have a responsibility to protect the environment.

We can protect our environment by:

- Planting trees
- Keeping our surroundings clean
- Avoiding pollution
- Protecting plants and animals

When the environment is healthy, living things can grow and survive.

Look Around!

Observe your surroundings. Choose one plant and one animal.

Discuss:

Plant:

- Where does it live?
- What does it need to survive?
- How does it help other living things?

Animal:

- Where does it live?
- What does it eat?

How does its environment help it survive?

Activity: Let Us Explore Habitats

Work with your classmates and identify different habitats around your community. Look at the examples of habitats in this lesson.

Discuss:

- Where does each living thing live?
- What does it get from its environment?
- Why do plants and animals need a suitable environment?
- What happens when their environment is damaged?

Share your ideas with the class.

Remember!

Living things live in different environments called habitats.

Examples of habitats include:

- Forests
- Farms
- Water bodies
- Grasslands
- Homes and communities

Living things depend on their surroundings for:

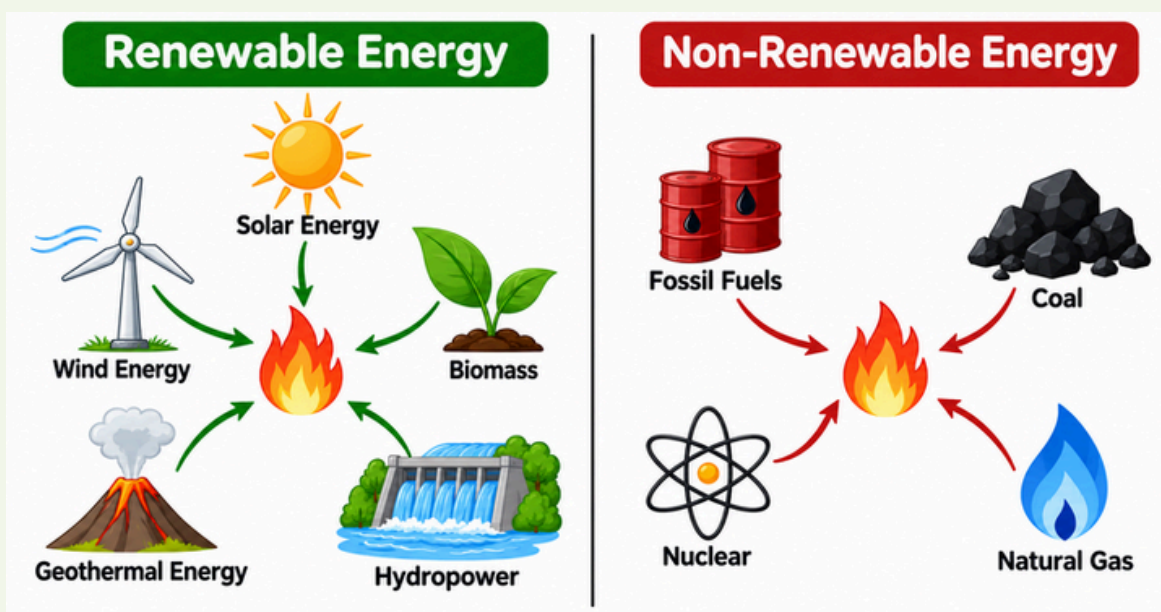
- Food
- Water
- Air
- Shelter

We must protect our environment because plants, animals and humans all depend on it to survive.

STRAND 4: DIVERSITY OF MATTER

UNIT 8: SOURCES AND FORMS OF ENERGY

LESSON 1: ENERGY AROUND US



Energy is an important part of our everyday lives. We need energy to do many activities such as walking, playing, cooking, lighting our homes and using machines.

Energy helps things happen and allows objects to move, work and change. Human beings, animals and machines all need energy to perform different activities.

We get energy from different sources around us. Some sources of energy come from nature, while others are made or stored by people.

Examples of things that need energy include:

- A person running
- A car moving
- A fan turning
- A lamp producing light
- A cooker heating food

Without energy, many activities around us would not happen.

Sources of Energy

A source of energy is something that provides energy for use. There are many sources of energy around us.

Some common sources of energy include:

- The Sun
- Food
- Electricity
- Fuel
- Batteries
- Wind

The Sun

The Sun is one of the most important sources of energy. It gives us light and heat.

Plants use energy from the Sun to make their own food and grow. Human beings and animals also depend on the Sun because it provides warmth and helps us see during the day.

We use energy from the Sun for activities such as:

- Drying clothes
- Drying crops
- Providing light during the day

The Sun is a natural source of energy that is available around us.

Food

Food is a source of energy for human beings and animals. When we eat food, our bodies get energy to help us:

- Walk
- Run
- Play
- Work
- Learn

Different foods provide energy in different amounts. Foods such as rice, maize, yam and bread help give our bodies the energy we need for daily activities. Animals also get energy from the food they eat so they can move and grow.

Electricity

Electricity is a form of energy that helps many devices and machines work.

We use electricity in our homes, schools and communities.

Examples of things that use electricity include:

- Light bulbs
- Fans
- Televisions
- Refrigerators

Electricity can come from different sources and is carried through wires to where it is needed.

We should use electricity safely by avoiding touching electrical appliances with wet hands and not playing with electrical sockets.

Fuel

Fuel is a source of energy that is used to produce heat and power machines.

Examples of fuels include:

- Petrol
- Diesel
- Gas
- Firewood

People use fuel for activities such as:

- Cooking
- Transporting people and goods
- Operating machines

Fuel should be used carefully because some fuels can produce smoke that affects the environment.

Batteries

Batteries store energy and provide power for small devices.

Examples of things that use batteries include:

- Torches
- Toys
- Clocks
- Remote controls

Batteries allow these devices to work even when they are not connected to electricity.

Wind

Wind is moving air that can provide energy. People use wind energy for activities such as moving sailing boats and turning wind machines. Wind is a natural source of energy that does not finish when it is used.

Uses of Energy

Energy is useful because it helps us carry out many activities.

Some important uses of energy include:

Providing Light

Energy helps us produce light.

Examples:

- Electricity powers light bulbs.
- The Sun provides natural light during the day.

Light helps us see and carry out activities.

Producing Heat

Energy helps us produce heat for different purposes.

Examples:

- Cooking food
- Heating water
- Drying clothes

The Sun, firewood, gas and electricity can provide heat.

Movement

Energy helps people, animals and machines move.

Examples:

- People use energy from food to walk and run.
- Cars use fuel energy to move.
- Machines use electricity to work.

Cooking

We use energy to prepare food.

Examples:

- Gas cookers
- Electric cookers
- Firewood stoves

Cooking makes food safe and easier to eat.

Using Machines and Devices

Many machines need energy to work.

Examples:

- Computers
- Fans
- Washing machines
- Refrigerators

Energy allows these devices to perform useful tasks.

Look Around!

Look around your home or classroom. Find things that use energy.

Discuss:

- What is the object?
- What source of energy does it use?
- What does the energy help it do?

Example:

Torch

- Source of energy: Battery
- Use: Produces light

Activity: Let Us Identify Energy Sources

Look around your home or classroom. Identify things that need energy to work.

Discuss:

- What is the object?
- What source of energy does it use?
- What does the energy help it do?
- Which sources of energy come from nature?
- Which objects need electricity?
- Why is energy important?

Share your ideas with your classmates

Remember!

Energy helps living things and machines work. Common sources of energy include:

- The Sun
- Food
- Electricity
- Fuel
- Batteries
- Wind

We use energy for:

- Light
- Heat
- Movement
- Cooking
- Operating machines

We should use energy wisely and safely because energy is important in our daily lives.

UNIT 9: ELECTRICITY AND ELECTRONICS

LESSON 1: SIMPLE ELECTRICAL DEVICES



Electricity is a form of energy that helps many objects around us work. We use electricity every day in our homes, schools and communities. Many devices need electrical energy to produce light, sound, heat or movement. These devices are called electrical devices.

Some electrical devices are simple and easy to understand. Examples include batteries, bulbs and electrical appliances. Learning about simple electrical devices helps us understand how the things we use everyday work and how to use them safely.

Batteries

A battery is a device that stores electrical energy. It provides energy to make some electrical devices work.

Batteries are small and can be used in many objects around us. Some devices cannot work without batteries because the batteries provide the energy they need.

Examples of devices that use batteries include:

- Torches
- Wall clocks
- Toys
- Remote controls
- Radios

When a battery is placed correctly inside a device, it supplies energy that allows the device to work.

For example, a torch uses a battery to produce light. Without the battery, the torch cannot shine. Batteries come in different sizes and shapes. Some batteries can be used once and thrown away, while others can be charged and used many times.

We should use batteries carefully. We should not break, throw into fire or play with batteries because they may be harmful.

Look Around!

Look around your home or classroom.
Can you find things that use batteries?

Discuss:

- What is the name of the device?
- Where is the battery placed?
- What does the battery help the device do?

Bulbs

A bulb is an electrical device that produces light when electrical energy passes through it. Bulbs help us see when it is dark. They are used in homes, schools, offices and other places.

Examples of places where bulbs are used include:

- Classrooms
- Bedrooms
- Streets
- Shops

A bulb needs a source of electrical energy to produce light. Some bulbs get energy from electricity supplied through wires, while others get energy from batteries.

There are different types of bulbs. Some bulbs are small, while others are large. Some are used inside buildings, while others are used outside. We should handle bulbs carefully because some bulbs can become hot when they are working.

Look Around!

Observe the bulbs around your school or home.

Discuss:

- Where are the bulbs located?
- Why are bulbs useful?
- What happens when there is no light.

Electrical Appliances

Electrical appliances are devices that use electricity to perform different tasks.

Many electrical appliances make our daily activities easier and faster.

Examples of electrical appliances include:

- Fans: uses power to spin its blades and create moving air that helps people feel cool.

- Televisions: receives electrical signals and converts them into visual displays and audio sounds that provide information and entertainment.
- Refrigerators: uses electricity to make things cold and helps keep food and drinks fresh.
- Irons: An iron uses electricity to produce heat that helps remove wrinkles from clothes.
- Computers: uses electricity to process information, show pictures and words, and help us do different tasks.
- Electric cookers: An electric cooker uses electricity to produce heat that helps us cook food.

Electrical appliances are useful because they help us do many tasks easily, but we must use them safely to prevent accidents and stay protected.

Safety Rules When Using Electrical Devices

Electricity is helpful in many ways, but it can be dangerous if we do not use it carefully and safely.

We should follow these safety rules:

- Do not touch electrical sockets with wet hands.
- Do not put fingers or objects into sockets.
- Do not play with electrical wires.
- Ask an adult for help when using electrical devices.
- Switch off devices when they are not needed.

Using electrical devices safely helps us avoid accidents and keeps us safe from harm.

Look Around!

Look around your classroom or home.

Identify three electrical devices.

Discuss:

- What is the name of the device?
- Does it use a battery or electricity from a socket?
- What does it help people do?

Activity: Let Us Identify Electrical Devices

Look around your classroom or home and identify electrical devices. With your teacher.

Discuss:

- What is the name of the device?
- Does it use a battery or electricity from a socket?
- What does the device help people do?
- How should we use the device safely?

Share your answer with your classmates.

Activity: Draw Simple Electrical Devices

Draw and label these devices:

- Battery
- Bulb
- Torch
- Fan

Share your drawings with your classmates.

Remember!

Simple electrical devices use electrical energy to work. They change electrical energy into other forms of energy, such as light, heat, sound, or movement, to perform useful tasks. Examples include batteries, bulbs, electrical appliances and so on.

Batteries

- Store electrical energy.
- Power devices such as torches and toys.

Bulbs

- Produce light.
- Help us see in dark places.

Electrical appliances

- Use electricity to perform different tasks.
- Examples include fans, televisions and refrigerators.

We should always use electrical devices safely.

UNIT 10: FORCES AND MOTION

LESSON 1: MOVEMENT AROUND US



Everything around us can be observed in different ways. Some things stay in one place, while others move from one place to another. Movement happens when an object changes its position.

We see movement everywhere around us. People walk and run, animals move from place to place, vehicles travel on roads, and objects can move when we push or pull them.

A force is a push or pull that can make an object move, stop, change direction or change shape. We use forces in our everyday activities. When we push, pull, kick, or move objects, we are applying force to make them move or change position.

Pushes and Pulls

A push is a force that moves an object away from us. When we push something, we use force to move it forward or away.

Examples of pushing include:

- Pushing a door open
- Pushing a chair closer to a table
- Pushing a toy car
- Pushing a box across the floor

When we push an object, the direction of the movement is away from our body. For example, when you push a ball, the ball moves away from you.

A pull is a force that moves an object towards us. When we pull something, we use force to bring it closer.

Examples of pulling include:

- Pulling a drawer open
- Pulling a bucket from a well
- Pulling a rope
- Pulling a bag towards you

When we pull an object, the object moves in the direction of the force. For example, when you pull a toy car with a string, the toy car moves towards you.

Push and Pull in Everyday Activities

We use pushes and pull in many activities around us.

Examples:

Opening a door:

- Pushing the door moves it away.
- Pulling the door brings it closer.

Playing games:

- Kicking a ball pushes it away.
- Pulling a rope during a game brings it closer.

Moving furniture:

- Pushing a table moves it forward.
- Pulling a chair brings it towards us.

Pushes and pulls help us move objects and perform tasks easily.

Look Around!

Look around your classroom.

Find objects that can be moved by pushing or pulling.

Discuss:

- What is the object?
- Do you push or pull it?
- What happens when you apply force?

Share your ideas with your classmates

Movement of Objects

Objects move in different ways depending on the force applied to them and the surface they move on. Some objects move quickly, while others move slowly. Some objects move in a straight line, while others move in different directions.

Examples of movement include:

- A ball rolling on the ground
- A car moving along a road
- A bird flying in the sky
- A person walking
- A swing moving back and forth

Ways Objects Move

Objects can move in different ways. They can move by rolling, sliding, spinning, moving back and forth, or travelling through the air. The way an object moves depends on the force applied to it.

Rolling Movement

Rolling happens when an object moves by turning over and over. It occurs when round objects, like balls and wheels, move along a surface.

Examples:

- A football rolling on the ground
- A wheel moving on a road
- A ball rolling down a slope

Objects with round shapes often roll easily because they can turn smoothly as they move along a surface.

Sliding Movement

Sliding happens when an object moves across a surface without rolling. It occurs when an object is pushed or pulled and moves from one place to another along a flat surface.

Examples:

- A book sliding across a table
- A box sliding across the floor

Some objects slide better on smooth surfaces than rough surfaces because there is less friction to slow their movement.

Moving Back and Forth

Some objects move repeatedly in two directions. This type of movement is called back-and-forth movement, like the motion of a swing or a clock pendulum.

Examples:

- A swing moving forward and backward
- A pendulum moving from side to side

Moving Through the Air

Some objects can move through the air. They can travel from one place to another when a force pushes or throws them, like a ball, a bird, or a flying kite.

Examples:

- Birds flying
- Kites flying
- Balls being thrown

Things That Affect Movement

Different things can affect how objects move. The amount of force, the shape of an object, and the surface it moves on can change its speed and direction.

Size and Weight

Heavy objects may need more force to move than lighter objects because they are harder to push or pull.

For example:

- A large box is harder to push than a small toy.

Surface

The surface an object moves on can affect its movement. A toy car moves more easily on a smooth floor than on rough ground because there is less friction.

Force Applied

The amount of push or pull affects movement. A stronger push or pull can make an object move faster or travel farther.

Look Around!

Observe objects moving around you. Notice how different objects move and identify whether they roll, slide, move back and forth, or travel through the air.

Discuss:

- How does the object move?
- Is it rolling, sliding or flying?
- What force makes it move?

Share your answers with your classmates.

Activity: Let Us Explore Movement

Work with your classmates. Look around your classroom and observe objects that can move. Try moving some objects carefully by using a push or pull, and describe how each object moves.

Discuss:

- What object did you move?
- Did you push or pull the object?
- How did the object move?
- Did it roll, slide, move back and forth, or move through the air?
- What force helped the object move?

Share your observations with your classmates.

Remember!

Movement happens when an object changes its position.

A force can be a:

Push

- Moves an object away from us.

Pull

- Moves an object towards us.

Objects can move by:

- Rolling
- Sliding
- Flying
- Moving back and forth

We use pushes and pulls every day to move objects around us.

STRAND 5: HUMANS AND OUR ENVIRONMENT

UNIT 11: HEALTH AND HYGIENE

LESSON 1: KEEPING OUR BODIES CLEAN



Our bodies are important, and we must take good care of them to stay healthy. Keeping our bodies clean helps us feel comfortable, look neat and prevent illnesses.

The practice of keeping our bodies clean is called personal hygiene. Personal hygiene involves the daily activities we do to keep different parts of our bodies clean and healthy.

Good hygiene habits help remove dirt and germs that can make us sick. Germs are tiny organisms that we cannot see with our eyes, but some of them can cause diseases. We should learn to take care of our bodies every day by practising good hygiene habits.

Personal Hygiene

Personal hygiene means keeping our body clean and taking care of ourselves. There are many ways we can practise personal hygiene.

Bathing Regularly

Bathing helps remove sweat, dirt and germs from our skin. We should bathe regularly using clean water and soap. Bathing keeps our body fresh and prevents unpleasant smells.

Keeping Our Hands Clean

Our hands touch many things every day. They can collect dirt and germs from objects around us.

We should wash our hands:

- Before eating
- After using the toilet
- After playing outside
- After touching dirty objects

Washing hands with clean water and soap helps prevent germs from entering our bodies.

Brushing Our Teeth

Our teeth help us chew food, so we must take care of them. We should brush our teeth every day, especially in the morning and before going to bed. Keeping our teeth clean helps prevent tooth problems and keeps our mouth healthy.

Keeping Our Hair Clean

We should keep our hair clean by washing and combing it regularly. Clean hair helps us look neat and prevents problems caused by dirt.

Wearing Clean Clothes

Clean clothes help protect our bodies and keep us comfortable. We should change dirty clothes, wash them properly and keep them neat.

Keeping Our Nails Clean

Dirt and germs can hide under our fingernails. Cleaning and trimming our nails regularly helps stop germs from spreading and keeps our bodies healthy.

We should:

- Trim our nails regularly
- Keep them clean
- Avoid biting our nails

Clean nails help prevent germs from entering our bodies. Keeping our nails neat and clean protects us from illnesses and helps us stay healthy.

Look Around!

Think about your daily activities. Discuss with your classmates:

- When do you wash your hands?
- How often do you brush your teeth?
- Why is bathing important?
- Why should we wear clean clothes?

Healthy Habits

Healthy habits are actions we practise regularly to keep our bodies strong and healthy. Good health does not only depend on keeping our bodies clean. We must also make good choices every day.

Some healthy habits include:

Eating Healthy Food

Food gives our bodies energy and helps us grow. It provides the nutrients we need to stay strong, healthy, and active.

We should eat different types of healthy foods, including:

- Fruits
- Vegetables
- Grains
- Meat
- Fish
- Eggs

Eating healthy food helps our bodies become strong. It gives us the energy to play, learn, and carry out our daily activities.

Drinking Clean Water

Our bodies need water to work properly. We should drink clean water every day and avoid drinking dirty water because it may contain germs that can make us sick.

Getting Enough Rest

Our bodies need time to rest after work and play. Children should get enough sleep every day so they can grow, learn and stay active.

Exercising and Playing

Physical activities such as running, walking and playing games help keep our bodies strong. Exercise helps our muscles and body parts work well.

Keeping Our Environment Clean

A clean environment helps keep us healthy. Keeping our surroundings clean prevents the spread of germs and helps us stay safe and well.

We should:

- Throw rubbish in bins
- Keep our homes and schools clean
- Avoid blocking drains
- Protect water sources

A clean environment reduces the spread of diseases. Keeping our homes, schools, and surroundings clean helps prevent germs from spreading and keeps us healthy.

Look Around!

Observe your school environment.

Discuss:

- Is your classroom clean?

- Where should rubbish be placed?
- What can you do to keep your school clean?

Activity: My Healthy Habits Chart

Think about the things you do every day to keep your body healthy.

Discuss with your classmates:

- When do you brush your teeth?
- When do you wash your hands?
- How do you keep your clothes clean?
- How do you keep your classroom clean?
- Why are healthy habits important?

Share your ideas with your classmates.

Activity: Let Us Demonstrate

Work with your classmates.

Demonstrate the correct way to:

- Wash your hands
- Brush your teeth
- Keep your classroom clean

Explain why each habit is important.

Remember!

Keeping our bodies clean helps us stay healthy.

Personal hygiene includes:

- Bathing regularly
- Washing hands
- Brushing teeth
- Keeping hair and nails clean
- Wearing clean clothes

Healthy habits include:

- Eating healthy food
- Drinking clean water
- Exercising
- Getting enough rest
- Keeping our environment clean

Taking care of our bodies every day helps us grow strong and healthy.

UNIT 12: DISEASES OF HUMANS

LESSON 1: UNDERSTANDING ILLNESSES



Our bodies need food, water, air and proper care to stay healthy. However, sometimes people become sick and do not feel well. Being sick is called an illness or disease.

An illness happens when something affects the normal way our body works. When we are sick, we may feel weak, tired or uncomfortable. Some illnesses can spread from one person to another, while others cannot.

We can learn to protect ourselves by understanding common diseases, their causes and how to prevent them.

Common Diseases

There are many diseases that affect human beings. Some diseases are common among children and adults.

Common diseases include:

- Malaria
- Cough and cold
- Diarrhoea
- Skin diseases
- Tooth decay

Let us learn about some of these diseases.

Malaria

Malaria is a common disease caused by germs carried by infected mosquitoes. When a mosquito carrying malaria germs bites a person, the person may become sick.

Common signs of malaria include:

- Fever
- Headache
- Body weakness
- Feeling cold
- Tiredness

We can reduce the risk of malaria by protecting ourselves from mosquito bites.

Cough and Cold

Cough and cold are common illnesses that affect our breathing system.

A person with a cough may:

- Cough often
- Have a sore throat
- Find it difficult to breathe comfortably

Cough and cold can spread when infected people release germs into the air through coughing or sneezing. We should cover our mouths when coughing or sneezing and keep our surroundings clean.

Diarrhoea

Diarrhoea is a sickness that causes a person to pass loose or watery stool frequently. It can be caused by eating unclean food or drinking unsafe water.

Signs of diarrhoea include:

- Passing watery stool often
- Stomach pain or discomfort
- Feeling weak or tired

Drinking safe water, eating clean food, and washing hands regularly can help prevent diarrhoea.

Skin Diseases

Skin diseases are illnesses that affect the skin. They can cause itching, rashes, or discomfort on different parts of the body.

Skin diseases can happen because of:

- Poor personal hygiene
- Dirty surroundings
- Contact with infected people or objects

Keeping our bodies clean and practising good hygiene helps protect our skin and keep it healthy.

Tooth Decay

Tooth decay is a problem that affects our teeth. It can happen when we do not brush our teeth properly or eat too many sugary foods. Keeping our teeth clean by brushing regularly helps prevent tooth decay.

Signs of tooth decay may include:

- Tooth pain
- Holes in teeth
- Difficulty eating

Brushing our teeth regularly helps keep our teeth healthy. It removes dirt and germs from our teeth and helps prevent tooth decay.

Look Around!

Think about times when you or someone you know was sick.

Discuss:

- What signs showed that the person was sick?
- What did the person do to get better?
- How can we avoid becoming sick?

Preventing Diseases

Many diseases can be prevented by practising healthy habits and taking good care of our bodies and surroundings. There are different ways we can help prevent diseases.

Keeping Our Bodies Clean

Keeping our bodies clean helps remove dirt and germs that can cause illness.

We should:

- Bathe regularly
- Wash our hands with soap and clean water
- Brush our teeth daily
- Wear clean clothes

Clean bodies help protect us from some diseases and keep us healthy.

Drinking Clean Water

Dirty water may contain germs that can cause illness.

We should:

- Drink clean water
- Keep water containers covered
- Avoid drinking water from unsafe sources

Clean water helps keep our bodies healthy.

Eating Clean and Healthy Food

Food should be prepared and stored properly to keep it safe and prevent germs from causing illness.

We should:

- Wash fruits and vegetables before eating
- Cover food to protect it from flies
- Eat balanced meals

Clean food helps prevent diseases. Eating fresh and properly prepared food protects our bodies from harmful germs and keeps us healthy.

Keeping Our Environment Clean

A dirty environment can allow germs and disease-causing insects to grow and spread.

We should:

- Dispose of rubbish properly
- Keep our surroundings clean
- Keep drains clear
- Avoid stagnant water

A clean environment helps protect us and the people around us from diseases.

Protecting Ourselves from Mosquitoes

We can help prevent malaria by reducing mosquito bites and keeping mosquitoes away.

We should:

- Sleep under mosquito nets
- Remove stagnant water around our homes
- Keep our surroundings clean

Taking these steps helps keep us safe and healthy.

Visiting Health Workers

When we feel sick, we should tell an adult and visit a health worker for help. Health workers can help identify illnesses and provide proper treatment. We should not take medicines without guidance from a responsible adult or health worker.

Look Around!

Observe your home or school environment.

Discuss:

- Are there places where germs can grow?
- Is there stagnant water where mosquitoes can breed?
- How can you help keep your surroundings clean?

Activity: Staying Healthy

Work with your classmates. Discuss common illnesses and how people can prevent them.

Talk about:

- What causes some diseases?
- How can we prevent diseases?
- Why is cleanliness important?
- Why should we protect our surroundings?

Share your ideas with the class.

Activity: Healthy Habits Role Play

Work in groups. Act out healthy actions such as:

- Washing hands
- Brushing teeth

- Cleaning the environment
- Covering your mouth when coughing

Explain how each action helps prevent diseases.

Remember!

Diseases can make our bodies weak and unhealthy. Practising good hygiene and healthy habits helps protect us from illnesses and keeps our bodies strong.

Common diseases include:

- Malaria
- Cough and cold
- Diarrhoea
- Skin diseases
- Tooth decay

We can prevent diseases by:

- Keeping our bodies clean
- Drinking clean water
- Eating healthy food
- Keeping our environment clean
- Seeking help when we are sick

Taking care of ourselves helps us stay healthy and strong. Eating healthy food, keeping clean, and practising good habits help our bodies grow and stay active.

UNIT 13: TECHNOLOGY IN LOCAL INDUSTRIES

LESSON 1: TECHNOLOGY AROUND US



Technology is part of our everyday lives. We use technology when we use tools, machines and devices that help us perform tasks more easily.

Technology means using knowledge, skills and materials to create things that solve problems and make work easier. People use technology in their homes, schools, farms, workplaces and communities.

Long ago, people used simple tools made from materials found in their environment. Today, people use both simple and advanced tools to help them complete different activities.

We can find technology around us in many places. In our homes, we use tools for cooking, cleaning and repairing things. In our communities, people use tools for farming, building, transportation and other activities.

Tools Used in Our Homes and Communities

A tool is an object that helps us perform a task. Tools make work easier, faster and safer. Different tools are used for different purposes. Some tools are used in homes, while others are used in communities.

Tools Used at Home

Many tools help us carry out activities in our homes. These tools make our daily tasks easier, faster, and safer. We use them for cooking, cleaning, repairing, and carrying out other household activities.

Examples include:

Knife

A knife is a tool used for cutting things. It helps us cut food, peel fruits, and prepare meals easily.

People use knives for:

- Cutting food
- Peeling fruits
- Preparing meals

We must handle knives carefully because they can be sharp. Using knives safely helps prevent injuries and keeps us protected while working.

Spoon

A spoon is a tool used for eating and stirring food. It helps us serve, mix, and enjoy different types of meals.

It helps us:

- Eat meals
- Mix ingredients when cooking

Broom

A broom is a tool used for cleaning. It helps us sweep floors, remove dirt, and keep our homes and surroundings neat.

People use brooms to:

- Sweep floors
- Remove dirt
- Keep surroundings clean

Hammer

A hammer is a tool used for hitting objects. It helps us drive nails into wood, repair items, and build different structures.

People use hammers to:

- Drive nails into wood
- Repair furniture
- Build structures

Bucket

A bucket is a container used for carrying and storing materials. It helps us carry water, collect items, and perform different household activities.

People use buckets for:

- Carrying water
- Washing
- Collecting items

Tools Used in Our Communities

People in communities use different tools to perform different jobs. These tools help farmers, builders, traders, and other workers complete their tasks more easily and efficiently.

Farming Tools

Farmers use tools to prepare land, plant crops, and harvest food. These tools help them work more easily and improve farming activities.

Examples:

- Hoe
- Wheelbarrow
- Rake
- Watering can
- Spade

These tools help farmers work on farms. They make farming activities easier, faster, and more effective.

Technology in Farming

Farmers use both simple tools and machines to make farming easier. Traditional tools like hoes, cutlasses, rakes, and watering cans help farmers prepare the land, plant crops, and care for plants. Some farmers also use machines like tractors to plough large areas of land quickly.

Technology helps farmers save time and produce more food for people in the community.

Building Tools

Builders use tools to construct houses, roads, and other structures. These tools help them measure, join, and build materials safely and efficiently.

Examples:

- Hammer
- Shovel
- Measuring tape
- Saw
- Trowel

These tools help builders complete their work. They make building activities easier, faster, and safer.

Technology in Construction

Builders and other workers use tools and machines to construct houses, roads, and other structures. Tools such as hammers, shovels, measuring tapes, and wheelbarrows help them complete their work safely and easily.

Modern machines also help workers carry heavy materials and finish construction projects faster.

Transport Tools

Some tools and machines help people move from one place to another. They also help transport goods and make travelling easier in our communities.

Examples:

- Bicycle
- Wheelbarrow
- Cart
- Truck
- Boat
- Bus

They help people carry goods and travel.

Technology in Transportation

People use technology to move themselves and goods from one place to another. Bicycles, carts, motorcycles, and vehicles help farmers, traders, and other workers transport people and materials.

Transportation technology makes movement easier and helps local businesses grow.

Look Around!

Look around your home or school. Find three tools.

Discuss:

- What is the name of the tool?
- What is it used for?
- Who uses it?

Example:

Hoe

- Used by farmers
- Helps prepare the soil

Knife

- Used at home.
- Helps people cut food.

Bicycle

- Used in communities.
- Helps people travel.

Hammer

- Used during building.
- Helps fix and construct things.

Share your ideas with the class.

Simple Machines

A simple machine is a device that helps us do work more easily. Simple machines reduce the amount of effort needed to move or lift objects. People use simple machines every day, often without knowing it.

Examples of simple machines include:

- Lever
- Wheel and axle
- Pulley
- Inclined plane
- Wedge

Lever

A lever is a simple machine that helps move or lift objects using a bar or rod. It makes work easier by reducing the effort needed to move heavy objects.

Examples of levers include:

- Seesaw
- Bottle opener
- Crowbar

A lever helps us use less effort to move heavy objects. It allows us to lift or move things more easily.

Wheel and Axle

A wheel and axle is a simple machine made of a wheel attached to a rod. It helps objects move more easily by reducing the effort needed to push or pull them.

Examples include:

- Bicycle wheels
- Doorknobs
- Carts

Wheels help objects move more easily from one place to another. They reduce the effort needed to move heavy objects and make transportation easier.

Pulley

A pulley is a simple machine that uses a wheel and rope to lift objects. It helps people raise heavy loads with less effort and makes lifting tasks easier.

Examples include:

- Drawing water from a well
- Lifting heavy materials at construction sites

A pulley helps people lift heavy objects with less effort. It makes lifting easier by changing the direction of the force used.

Inclined Plane

An inclined plane is a sloping surface that helps move objects from one level to another. It makes it easier to lift or move heavy objects by reducing the amount of effort needed.

Examples include:

- Ramps
- Sloping roads

A ramp helps people move heavy objects more easily. It allows objects to be pushed or pulled from one level to another with less effort.

Wedge

A wedge is a simple machine with a sharp edge used for cutting or splitting objects. It helps make tasks like cutting wood, food, or other materials easier.

Examples include:

- Knife
- Axe
- Nail

A wedge helps make cutting easier. It uses its sharp edge to split or separate materials with less effort.

Technology Makes Work Easier

Technology helps people save time and energy. It makes tasks easier, faster, and more efficient by reducing the effort needed to complete different activities.

Examples:

- A wheelbarrow helps carry heavy loads.
- A bicycle helps people travel faster.
- A hammer helps fix objects.
- A knife helps cut materials.

Without tools and machines, many tasks would take more time and effort. Tools and machines help people complete their work faster, easier, and more safely.

Look Around!

Find examples of simple machines around you.

Discuss:

- What simple machine can you identify?
- What work does it help people do?
- How does it make work easier?

Activity: Let Us Explore Tools and Machines

Look around your home, school or community. Identify tools and machines people use.

Discuss:

- What is the name of the tool?
- Where is it used?
- What work does it help people do?
- Why is the tool useful?

Share your observations with your classmates.

Activity: Think Like an Inventor

Think of a task that is difficult to do. Imagine a simple tool or machine that can help make the task easier and explain how it works.

Discuss:

- What problem does the task create?
- What tool could make the work easier?

- How would the tool work?

Share your answer with your classmates

Remember!

Technology helps people solve problems and make work easier. It provides tools and machines that help us complete tasks more quickly and efficiently.

Tools

- Help us perform tasks.
- Examples: knife, broom, hammer, hoe.

Simple machines

- Reduce the effort needed to do work.

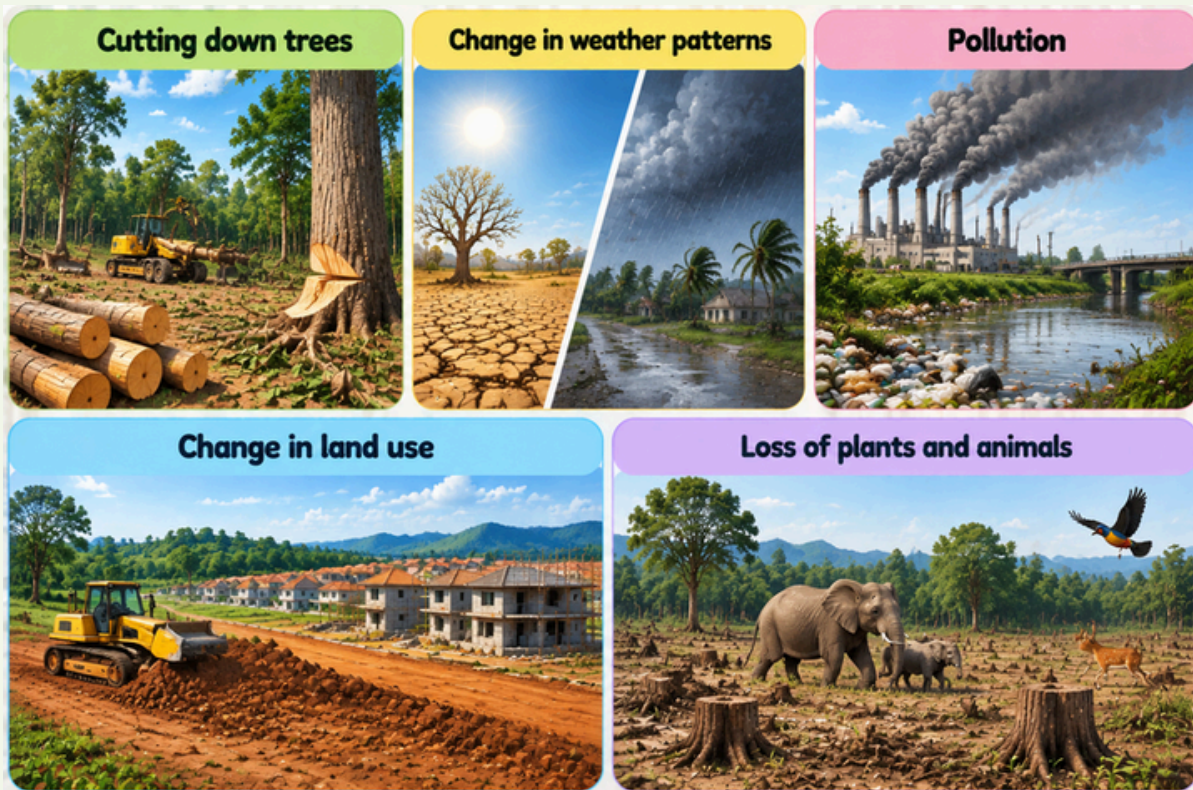
Examples:

- Lever
- Wheel and axle
- Pulley
- Inclined plane
- Wedge

We should use tools and machines safely and responsibly.

UNIT 14: CLIMATE CHANGE

LESSON 1: CHANGE IN OUR ENVIRONMENT



The environment is everything around us. It includes the air we breathe, the water we drink, the land we live on, plants, animals and all the things that surround us.

Human beings, plants and animals all depend on the environment to survive. We get food, water, shelter and materials from our environment. This is why it is important for everyone to take care of nature.

The environment does not always remain the same. It can change because of natural events and human activities. Some changes are helpful, while others can harm plants, animals and people.

By understanding changes in our environment and learning how to protect nature, we can help keep our world safe and healthy.

Changes in Our Environment

Our environment changes over time. Some changes happen naturally, while others happen because of the activities of people. We can observe changes in our surroundings by looking at plants, animals, weather conditions and the places where we live.

Examples of changes in our environment include:

- Cutting down trees changes in weather patterns
- Pollution
- Changes in land use
- Loss of plants and animals
-

Cutting Down Trees

Trees are important parts of our environment. They provide oxygen, shade, food and homes for many animals. When too many trees are cut down without planting new ones, the environment can be affected.

Cutting down many trees can cause:

- Loss of homes for animals
- Less shade
- Soil erosion
- Changes in weather conditions

We should plant and protect trees to keep our environment healthy. Trees provide clean air, shade, and help protect the environment for people, animals, and future generations.

Pollution

Pollution happens when harmful materials make our environment dirty. It can affect the air, water, and land, making it unsafe for people, plants, and animals.

Pollution can affect:

- Air
- Water
- Land

Examples of pollution include:

- Throwing rubbish on the ground
- Dumping waste into rivers
- Smoke from vehicles and burning materials

Pollution can make people, plants and animals sick. We can reduce pollution by keeping our surroundings clean and disposing of waste properly.

Changes in Weather

The weather around us can change. Sometimes we experience very heavy rainfall, very hot days or longer dry periods.

Changes in weather can affect:

- Farming
- Water supply
- Plants and animals
- Human activities

Observing weather changes helps us understand our environment and prepare for different conditions.

Look Around!

Look around your school or community. Observe your environment and discuss:

- Are there many trees around you?
- Is the environment clean or dirty?
- Can you see any signs of pollution?
- What changes have you noticed in your surroundings?

Share your observations with your classmates.

Protecting Nature

Nature includes all living things and natural resources around us. Protecting nature means caring for plants, animals, water, air, and land. Everyone has a responsibility to care for the environment. There are many ways we can protect and care for nature.

Planting Trees

Trees are important because they provide:

- Clean air
- Shade
- Food
- Homes for animals

Planting trees helps replace trees that have been removed and keeps the environment healthy. Trees provide clean air, give shade, and support the lives of people and animals.

Keeping Our Surroundings Clean

A clean environment helps people, plants, and animals stay healthy. It reduces pollution and provides a safe place for living things to grow and survive.

We can keep our surroundings clean by:

- Throwing rubbish into bins
- Cleaning our homes and schools
- Avoiding littering
- Recycling materials when possible

Protecting Water Sources

Water is important for all living things. People, plants, and animals need water to survive, grow, and stay healthy.

We can protect water by:

- Keeping rivers and streams clean
- Not throwing rubbish into water bodies
- Using water carefully

Clean water helps humans, animals, and plants survive. It provides the water needed for growth, health, and everyday life.

Look Around!

Look around your school or community. Observe your environment and discuss:

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- Is the environment clean or dirty?
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Protecting Plants and Animals

Plants and animals are important parts of nature. They help keep the environment balanced and provide many benefits to living things.

We can protect them by:

- Not destroying their habitats
- Caring for domestic animals
- Avoiding unnecessary killing of animals
- Planting more plants

When plants and animals are protected, the environment remains balanced. Protecting them helps maintain nature and allows living things to survive.

Reducing Waste

Waste can harm the environment when it is not properly managed. It can make our surroundings dirty and cause pollution that affects people, plants, and animals.

We can reduce waste by:

- Reusing materials
- Recycling items
- Avoiding unnecessary use of plastics
- Keeping waste in proper places

Small actions can make a big difference in protecting nature. Simple habits like keeping our surroundings clean, planting trees, and saving water help care for the environment.

Look Around!

Think about your daily activities. Identify the ways you can help protect nature and keep your environment clean.

Discuss:

- How do you help keep your home clean?
- What can you do to protect plants?
- How can you reduce waste?

Activity: Let Us Care for Our Environment

Work with your classmates. Walk around your school and observe the environment.

Look Around!

Observe your school or community.

Discuss:

- Are there trees around you?
- Is the environment clean?
- Can you see any rubbish or pollution?
- What changes have you noticed in your surroundings?
- What can people do to protect the environment?

Activity: My Environment Promise

Write one thing you will do to protect nature. For example, you can plant a tree, dispose of waste properly, or keep your surroundings clean.

Examples:

- I will plant trees.
- I will not throw rubbish on the ground.
- I will keep my classroom clean.
- I will protect animals.

Share your promise with your classmates.

Remember!

The environment is important because all living things depend on it. It provides the air, water, food, and resources needed for humans, plants, and animals to survive.

Environmental changes can happen because of:

- Human activities
- Natural events
- Pollution

We can protect nature by:

- Planting trees
- Keeping our surroundings clean
- Protecting water
- Caring for plants and animals
- Reducing waste

When we care for the environment, we protect our future. Keeping nature clean and safe helps create a healthy place for people, plants, and animals to live.

STRAND 6: THE FOOD SYSTEM

UNIT 11: THE FOOD WE EAT

LESSON: TYPES OF FOOD



Food is an important part of our daily lives. All living things need food to grow, stay healthy and get energy. Human beings eat different kinds of food from plants and animals.

The food we eat comes from different sources. Some foods come from plants, while others come from animals. We get food from farms, gardens, rivers, markets and other places around us.

The type of food we eat can affect how our bodies grow and how healthy we become. Eating different kinds of food helps our bodies get the nutrients they need.

Foods We Eat

There are many different types of food around us. The foods we eat depend on where we live, what is available and the traditions of our communities.

Some common foods we eat include:

- Maize
- Rice

- Yam
- Cassava
- Plantain
- Beans
- Fruits
- Vegetables
- Fish
- Meat
- Eggs
- Milk

These foods come from different sources. Some foods come from plants, while others come from animals or other natural sources.

Foods from Plants

Many foods come from plants. Farmers grow different crops that provide food for people and animals.

Examples of foods from plants include:

Maize

Maize is a common crop that is used to prepare different meals. It can be eaten as porridge, boiled maize or other food products.

Rice

Rice is a popular food eaten in many homes. It can be cooked and eaten with different types of soup or sauce.

Cassava

Cassava is a root crop that grows underground. It can be used to prepare foods such as gari and other meals.

Plantain

Plantain is a crop that grows on a plant. It can be boiled, fried or prepared in different ways.

Fruits and Vegetables

Fruits and vegetables come from plants. They provide our bodies with important nutrients that help us grow, stay healthy, and strong.

Examples include:

- Mango
- Orange
- Tomato
- Carrot
- Cabbage

Fruits and vegetables help keep our bodies healthy. They provide vitamins and nutrients that help us grow, stay strong, and have energy.

Foods from Animals

Some foods come from animals. Animals provide foods that help our bodies grow and become strong.

Examples of foods from animals include:

Meat

Meat comes from animals such as cows, goats and chickens. It provides nutrients that help our bodies grow.

Fish

Fish live in water and provide food for people. Fish can be obtained from rivers, lakes and oceans.

Eggs

Eggs are produced by birds such as chickens. They are eaten in many homes.

Milk

Milk comes from animals such as cows and goats. It helps provide nutrients needed by the body.

Animals are important because they provide food and other useful materials.

Different Types of Food

Foods can be grouped according to what they do in our bodies. Different groups of foods help us grow, provide energy, and protect our bodies from sickness.

Energy-Giving Foods

Energy-giving foods provide energy for our bodies to work, play, and learn. They help us stay active and carry out our daily activities.

Examples include:

- Rice
- Maize
- Yam
- Cassava
- Bread

These foods help us have strength for our daily activities. They give our bodies the energy needed to move, learn, play, and stay active.

Body-Building Foods

Body-building foods help our bodies grow and repair damaged parts. They help build strong muscles, bones, and other body tissues.

Examples include:

- Meat
- Fish
- Eggs
- Beans
- Milk

Children need body-building foods because their bodies are growing. These foods help them develop strong muscles, bones, and healthy body parts.

Protective Foods

Protective foods help protect our bodies from diseases and keep us healthy. They provide vitamins and minerals that help our bodies fight illnesses and stay strong.

Examples include:

- Fruits
- Vegetables

Eating fruits and vegetables helps our bodies fight illnesses. They provide important nutrients that keep us healthy and help our bodies work properly.

Importance of Food

Food is important because it helps us live, grow, and stay healthy. It gives our bodies energy, helps us grow strong, and protects us from illnesses.

Some important benefits of food include:

Food Gives Us Energy

Food provides energy that helps us perform daily activities. It gives us the strength to work, play, learn, and stay active.

We need energy to:

- Walk
- Run
- Play
- Work
- Learn

Without enough food, our bodies become weak. We may lack the energy needed to grow, learn, and carry out our daily activities.

Food Helps Us Grow

Food provides nutrients that help children grow properly. These nutrients help build strong bodies, support development, and keep children healthy.

Healthy food helps our:

- Bones grow strong
- Muscles develop
- Bodies become stronger

Food Keeps Us Healthy

Eating the right types of food helps protect our bodies from diseases. Fruits and vegetables provide vitamins that help keep us healthy.

Food Repairs Our Bodies

When our bodies are injured or damaged, some foods help repair and rebuild body tissues. Foods such as fish, eggs, meat and beans help our bodies repair themselves.

Eating a Balanced Diet

A balanced diet means eating different types of food in the right amounts. It provides the nutrients our bodies need to grow, stay healthy, and have energy.

A healthy meal should include:

- Energy-giving foods
- Body-building foods
- Protective foods

For example:

A meal containing rice, fish and vegetables provides different nutrients needed by the body. We should avoid eating too much of one type of food. Eating a variety of foods helps us stay healthy.

Look Around!

Look at the foods available in your home or school.

Discuss:

- What foods do you eat every day?
- Which foods come from plants?
- Which foods come from animals?
- Why do we need food?

Activity: Let Us Identify Foods

Work with your classmates. Look at different examples of foods in this lesson.

Discuss:

- What is the name of the food?
- Does it come from a plant or animal?
- Is it an energy-giving food?
- Is it a body-building food?
- Is it a protective food?

Examples:

Rice

- Comes from plants.
- Gives the body energy.

Fish

- Comes from animals.
- Helps the body grow.

Share your answers with the class.

Activity: Exploring Our Meals

Think about a meal you enjoy eating.

Discuss:

- What foods are included in the meal?
- Which foods come from plants?
- Which foods come from animals?
- How does the meal help your body?

Remember!

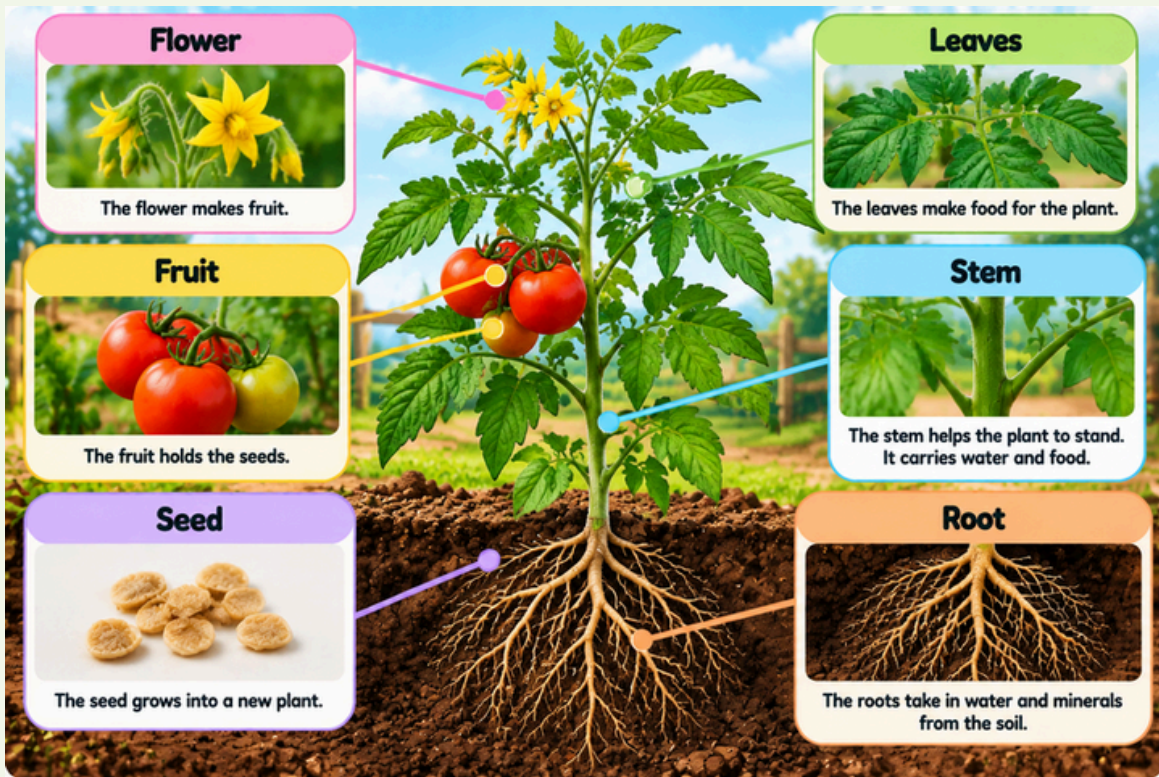
Food is important because it helps us:

- Get energy
- Grow strong
- Stay healthy
- Repair our bodies

Foods come from plants and animals. Plant foods include fruits, vegetables, and grains, while animal foods include meat, fish, eggs, and milk. Eating different types of food helps us grow healthy and strong.

UNIT 16: GARDENING

LESSON 1: GROWING PLANTS



Plants are important living things that are found around us. They provide food, clean air, shade and materials that people use every day. Many plants around us begin their lives from seeds. When a seed is planted in the right conditions, it can grow into a new plant.

Growing plants requires care and attention. Farmers, gardeners and other people help plants grow by providing water, sunlight and other things they need. Learning how plants grow helps us understand the importance of plants and how we can take care of them.

Parts of Plants

Plants have different parts that help them grow and survive. Each part of a plant has a special function.

The main parts of a plant include:

- Roots
- Stem
- Leaves
- Flowers
- Fruits
- Seeds

Roots

Roots are the parts of a plant that grow below the ground. They hold the plant firmly in the soil and absorb water and nutrients needed for growth.

Roots help the plant by:

- Holding the plant firmly in the soil
- Taking in water from the soil
- Absorbing nutrients needed for growth

For example, the roots of a maize plant hold it upright and help it get water from the soil.

Stem

The stem is the part of the plant that supports the leaves, flowers, and fruits. It also carries water and nutrients to different parts of the plant.

The stem helps the plant by:

- Holding the plant upright
- Carrying water to other parts of the plant
- Supporting leaves and flowers

Examples of plants with strong stems include trees and maize plants. Strong stems help these plants stand upright and support their leaves, flowers, and fruits.

Leaves

Leaves are usually green parts of plants that grow from the stem. They help plants make food and receive sunlight needed for growth.

Leaves help plants by:

- Making food for the plant using sunlight
- Taking in air needed for growth

Healthy leaves help plants grow well. They help plants make food and stay strong by using sunlight, air, and water.

Flowers

Flowers are the colourful parts of some plants. They help plants produce seeds for new plants to grow. Some flowers later develop into fruits that people and animals can eat.

Examples of plants with flowers include:

- Mango trees
- Tomato plants
- Sunflowers

Fruits and Seeds

Fruits protect seeds and help new plants grow. Seeds are the small parts of plants that can develop into new plants when they are planted in the soil and given the right conditions.

Examples of seeds include:

- Maize seeds
- Bean seeds
- Groundnut seeds

Look Around!

Observe a plant around your home or school.

Identify:

- The roots
- The stem
- The leaves
- The flowers or fruits

Discuss:

- Which part is above the soil?
- Which part is below the soil?
- What does each part do?

Planting Seeds

Plants can be grown by planting seeds. A seed contains a small baby plant that can develop into a bigger plant when it gets water, air, and sunlight.

For a seed to grow, it needs the right conditions. These include:

- Water
- Air
- Suitable soil
- Sunlight

The process of a seed beginning to grow is called germination. During germination, the seed starts to develop into a young plant when it receives water, air, and suitable conditions.

During germination:

- The seed absorbs water from the soil.
- A small root grows downward.
- A young stem grows upward.
- Leaves begin to appear.
- The young plant continues to grow.

Steps for Planting Seeds

Step 1: Prepare the Soil

The soil should be soft and suitable for planting. Remove stones and unwanted materials.

Step 2: Plant the Seed

Make a small hole in the soil and place the seed inside. Cover the seed gently with soil and leave it in a suitable place where it can begin to grow.

Step 3: Provide Water

Water the seed regularly to help it grow. Too much water can damage the seed, while too little water can prevent it from growing properly.

Step 4: Provide Sunlight

Place the plant where it can receive enough sunlight. Sunlight helps plants grow well and keeps them healthy.

Look Around!

Think about plants grown in your community.

Discuss:

- What crops do farmers plant?
- What seeds are used?
- What does the plant need after planting?

Caring for Plants

Plants need proper care to grow strong and healthy. When we take care of plants, they can produce food, flowers, and fruits that are useful to people and animals.

Ways to care for plants include:

Watering Plants

Plants need water to survive and grow. We should provide enough water, especially during dry periods, to help them remain healthy and continue growing.

Removing Weeds

Weeds are unwanted plants that grow near crops. They compete with crops for water, nutrients, and sunlight, so they should be removed to help crops grow well.

Weeds compete with plants for:

- Water
- Nutrients
- Space

Removing weeds helps plants grow better. It allows plants to get enough water, nutrients, and sunlight needed for healthy growth.

Providing Sunlight

Plants need sunlight to make their own food. We should plant crops in places where they can receive enough sunlight so they can grow well and remain healthy.

Protecting Plants from Damage

Plants can be damaged by animals, insects, or careless activities. Protecting plants from harm helps them grow well and produce healthy crops, flowers, and fruits.

We should protect plants by:

- Avoiding unnecessary cutting
- Keeping animals away from young plants

- Handling plants carefully

Adding Nutrients to the Soil

Plants need nutrients from the soil to grow well. Farmers may add compost or manure to improve the soil and provide the nutrients plants need for healthy growth.

Activity: Let Us Plant a Seed

Work with your teacher and classmates.

You will need:

- A seed
- Soil
- A container
- Water

Steps:

- Put soil into the container.
- Plant the seed in the soil.
- Add a small amount of water.
- Place the container where it can receive sunlight.
- Observe the seed every day and see if any changes occurs.

Share what you observe with your classmates.

Activity: Observe a Plant

Choose a plant around your school. Observe the different parts of the plant.

Discuss:

- Where are the roots found?
- Where is the stem found?
- What do the leaves do?
- Can you see flowers or fruits?
- Why are different plant parts important?

Share your observations with the class.

Remember!

Plants have different parts that help them grow. Each part has a special function that helps the plant survive and stay healthy.

Roots

- Hold the plant in the soil.
- Absorb water.

Stem

- Supports the plant.
- Carries water.

Leaves

- Help the plant make food.

Flowers and fruits

- Help plants produce seeds.

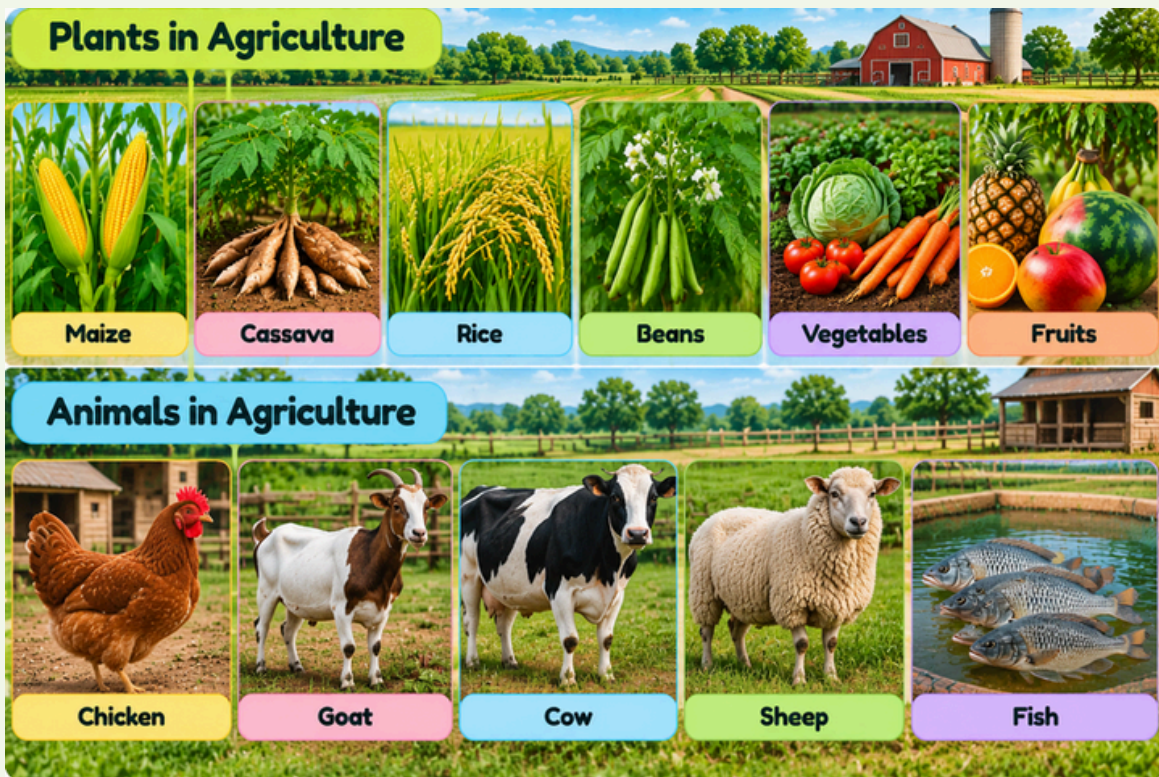
Plants grow from seeds and need:

- Water
- Air
- Soil
- Sunlight

We should care for plants because they are important to humans, animals and the environment.

UNIT 17: THE AGRICULTURE WORLD

LESSON 1: PLANTS AND ANIMAL IN AGRICULTURE



Agriculture is the practice of growing plants and keeping animals for food and other useful products. Agriculture is an important activity in many communities because it provides food, materials and jobs for people.

Farmers use land, water, tools and other resources to grow crops and take care of animals. The plants and animals raised by farmers are called agricultural products.

Plants and animals in agriculture help provide many things we need every day. The food we eat, the clothes we wear and some materials we use come from agriculture.

There are two important parts of agriculture:

- Growing crops
- Keeping farm animals

Farm Crops

Farm crops are plants that farmers grow for different purposes. Crops are grown on farms, gardens and other places where plants can receive the things they need to grow.

Crops need:

- Good soil
- Water
- Air
- Sunlight
- Proper care

Farmers prepare the land, plant seeds, remove weeds, and protect crops from damage. These activities help crops grow well and produce healthy food.

Some common farm crops include:

Maize

Maize is an important crop grown by many farmers. It is used as food for people and animals. Maize can be prepared in different ways and is eaten in many communities.

Cassava

Cassava is a root crop that grows underground. It is used to prepare different foods and provides energy for the body.

Rice

Rice is a common food crop grown in many places. Farmers grow rice because it provides food for many people.

Beans

Beans are crops that provide food and nutrients. They are often eaten with other foods such as rice, maize and other meals.

Vegetables

Vegetables are important crops that help keep our bodies healthy. Vegetables provide vitamins and other nutrients needed by the body.

Examples include:

- Tomato
- Cabbage
- Pepper
- Garden eggs

Fruits

Fruit crops provide food and help keep us healthy. Fruits contain nutrients that help protect our bodies.

Examples include:

- Mango
- Orange

- Banana
- Pineapple

Fruits contain nutrients that help protect our bodies

Look Around!

Visit a farm, garden or market with your teacher or parents. Observe different crops.

Discuss:

- What is the name of the crop?
- Where does it grow?
- How is it useful to people?

Farm Animals

Farm animals are animals that people keep for food, work, or other useful products. They provide things such as milk, eggs, meat, and help with some farm activities. Farmers provide animals with:

- Food
- Water
- Shelter
- Protection

Taking good care of farm animals helps them grow healthy. Some common farm animals include:

Chickens

Chickens are common farm animals. They are kept for their eggs and meat and are easy to raise on many farms.

They provide:

- Eggs
- Meat

Farmers keep chickens in safe places called poultry houses. These houses protect the chickens from bad weather, predators, and other dangers.

Goats

Goats are animals kept by farmers for different purposes. They can provide milk, meat, and other useful products.

They provide:

- Meat
- Milk
- Other useful products

Goats eat grass, leaves, and other plant materials. They use their strong mouths to chew and eat different types of plants.

Cows

Cows are large farm animals. They are raised for milk, meat, and other useful products.

They provide:

- Meat
- Milk

In some communities, cows may also help farmers with work on farms. They can be used to pull carts or help with other farm activities.

Sheep

Sheep are farm animals kept for:

- Meat
- Wool

They feed on grass and need proper care to stay healthy. They are also raised by farmers for breeding and other useful purposes.

Fish

Fish farming involves keeping and raising fish in ponds or other suitable water spaces. Fish provide food and are an important source of protein for people.

Look Around!

Think about animals kept in your community. Identify the farm animals you see and discuss how they help people.

Discuss:

- Which farm animals have you seen?
- Where do they live?
- What do they provide for people?

Importance of Agriculture

Agriculture is important because it provides many things that humans need. It gives us food, materials, and other products that support our daily lives.

Providing Food

Agriculture provides food for people and animals. Crops such as maize, rice, and vegetables provide meals for families, while animals such as chickens, goats, and fish provide food products like meat, eggs, and fish.

Providing Jobs

Many people work in agriculture. Farmers and other workers help grow crops, raise animals, and provide food for communities.

Examples include:

- Farmers
- Fishermen
- Animal keepers
- Food processors

Agriculture helps people earn income to support their families. Farmers can sell crops, animals, and other farm products to make money.

Providing Raw Materials

Agriculture provides materials used to make different products. Materials such as cotton, rubber, and plant fibres are used to produce items we use every day.

Examples:

- Cotton from plants is used to make fabric.
- Crops can be used to make food products.

Supporting Communities

Farming activities help communities develop. Farmers sell crops and animals in markets, providing food and other goods for people while supporting the local economy.

Protecting the Environment

Good farming practices can help protect nature. They help keep the soil healthy, reduce waste, and protect plants, animals, and the environment.

Farmers can care for the environment by:

- Planting trees
- Protecting soil
- Using water wisely

Look Around!

Think about your community. Identify the farming activities, crops, and animals that can be found around you.

Discuss:

- Are there farms near where you live?
- What crops do farmers grow?
- What animals do farmers keep?
- How does agriculture help your community?

Activity: Let Us Explore Agriculture

Work with your classmates. Discuss agricultural products found in your community. Talk about:

- What crops do farmers grow?
- What animals do farmers keep?
- How do these products help people?
- Why is agriculture important?

Examples:

Maize

- A crop grown by farmers.
- Used as food.

Chicken

- A farm animal.
- Provides eggs and meat.

Share your ideas with the class.

Activity: Exploring a Farm

Look at pictures of farms. Identify the crops, animals, tools, and activities you can see in each farm picture.

Discuss:

- What crops can be found on farms?
- What animals can be found on farms?
- What work do farmers do?
- How does farming help people?

Remember!

Agriculture involves growing crops and keeping animals. It provides food, materials, and other products that people need for daily life.

Farm crops include:

- Maize
- Cassava
- Rice
- Beans
- Vegetables
- Fruits

Farm animals include:

- Chickens
- Goats
- Cows

- Sheep
- Fish

Agriculture is important because it provides:

- Food for people and animals
- Materials used to make products
- Income for farmers and workers
- Support for communities and the economy

Farmers help provide the things we need to live healthy lives. They grow crops and raise animals that give us food and other useful products.